ACTUATORS



SERVO MOTOR

- Servo is mechanism based on feedback control.
  - A servo motor is a high-quality DC motor that qualifies to be used in a “servoing application”
- A motor with a "feedback" device
- Electronic packages control speed and position
- There are two type of servo motor
  - Standard servo motor .motor turn from 0 to 180°
  - Continuous servo motor .motor turn form 0 to 360°

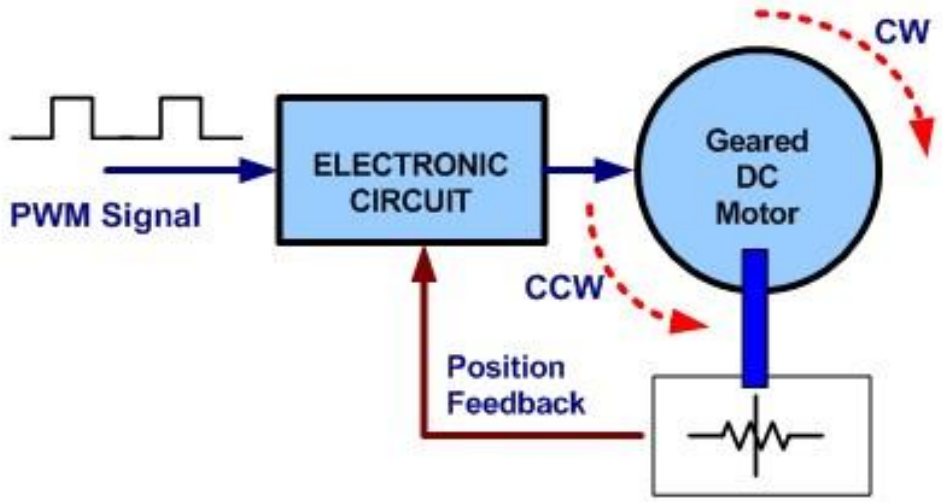


SERVOS

http://www.ermicro.com/blog



Typical Servo Motor

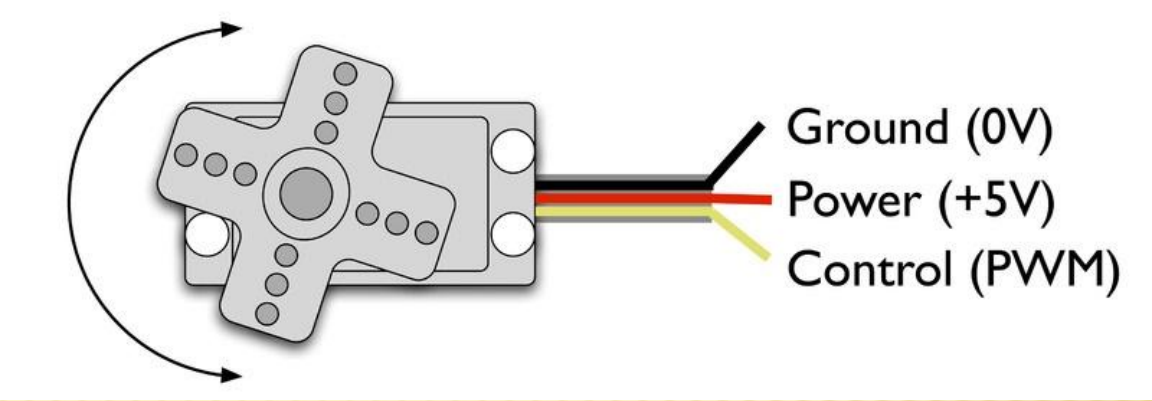


Servo Motor Block Diagram

The Servo Motor

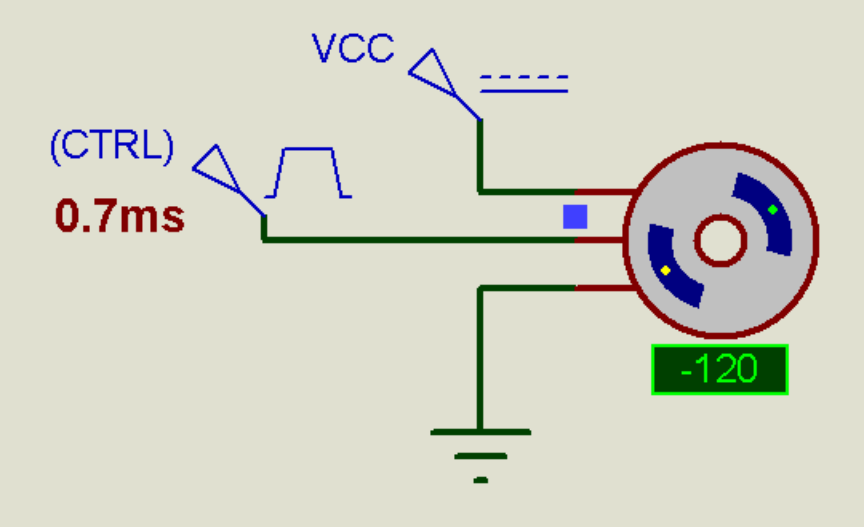
- A servo is a unit combining motor and simple feedback electronics for position control
- A servo is set by applying a PWM signal of a certain ratio
- Ratio determines servo position, not speed!
- Servos are usually used in model airplanes, etc.

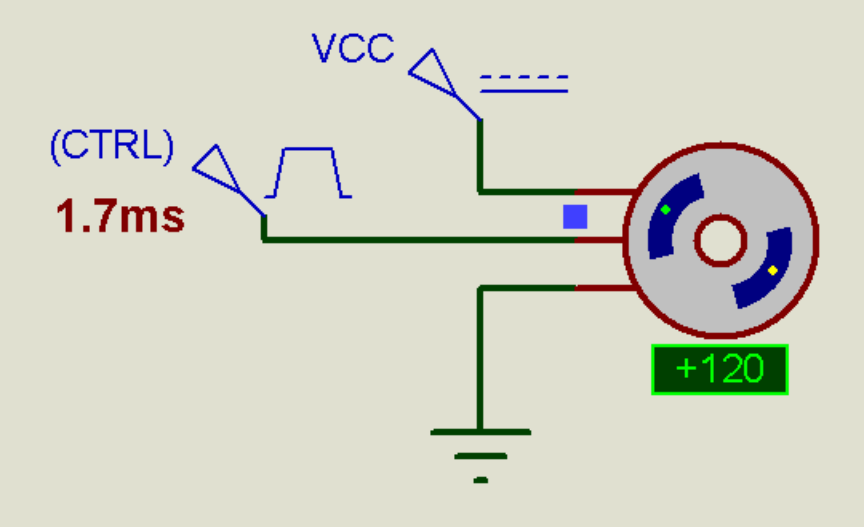
SERVOS

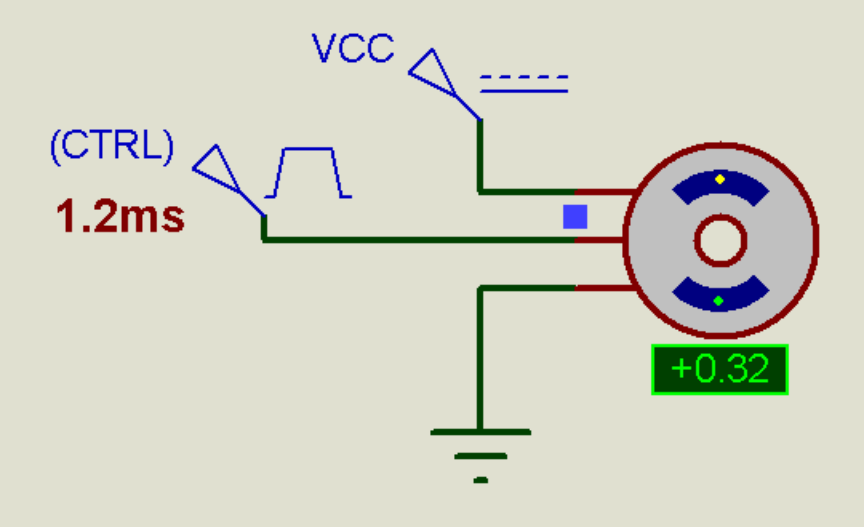


- Servo usually have three wires: power, ground and PWM signal
- The PWM signal frequency is 50Hz (T=20ms)
- The pulse width should be between 0.5ms and 2ms long
- For example, a width of 0.7ms will rotate the disk to the leftmost position (–120°), and a width of 1.7ms will rotate the disk to the rightmost position (+120°).

SERVOS







WHERE ARE SERVOS USED?

- The **Servos** are used for precision positioning. They are used in robotic arms and legs, sensor scanners and in RC toys like RC helicopter, airplanes and cars

- https://www.youtube.com/watch?v=B2fI8-L6xcA



SERVOS PROS & CONS

- **Servo strengths (adv):**
  1. High performance
  2. Small size
  3. Wide variety of components
  4. High speeds available with specialized controls
- **Servo weaknesses(dis adv):**
  1. Slightly higher cost
  2. High performance limited by controls

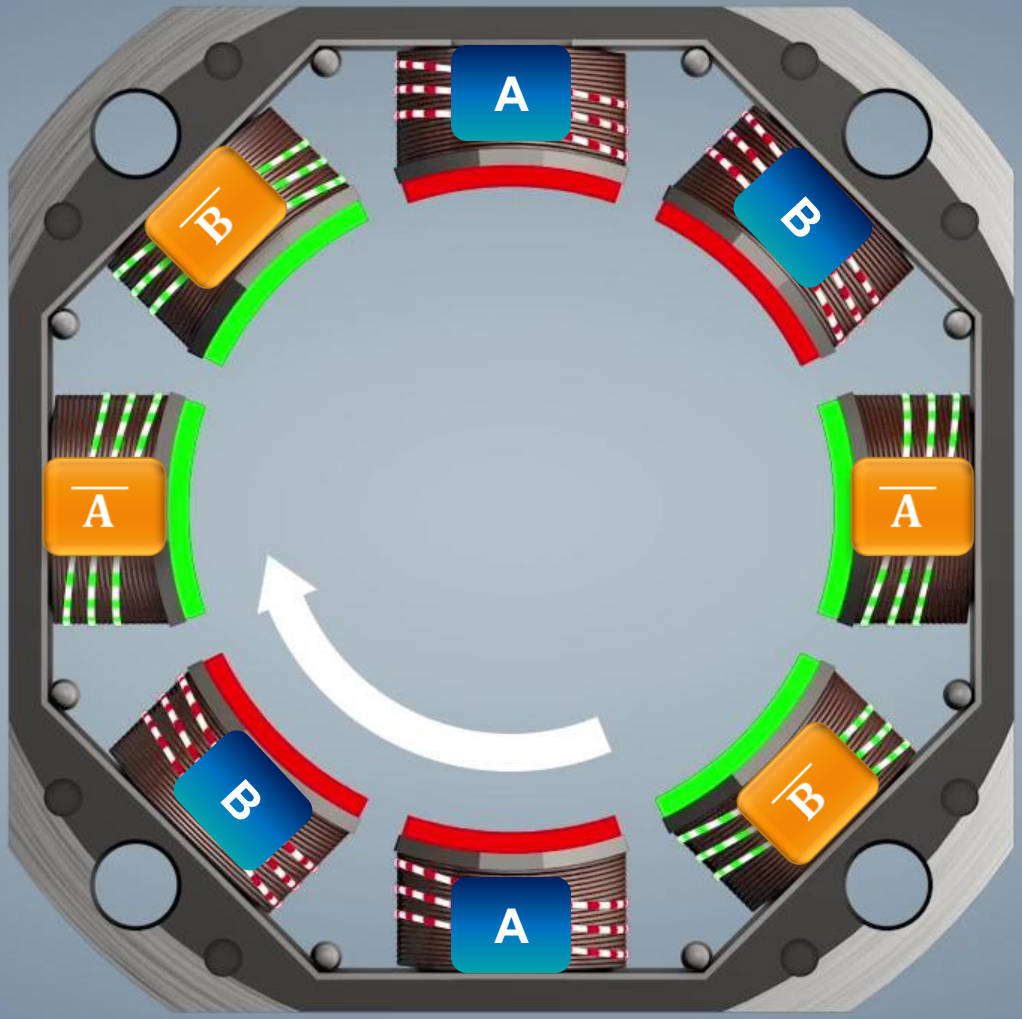


WHAT IS STEPPER MOTOR?

- A stepper motor or Stepping motor is a motor that, as the name suggests, moves in steps.
- It can perform position or speed control
- It is used in simple open-loop control systems; these are generally adequate for systems that operate at low accelerations with static loads,
- Can go forward or reverse (backward)

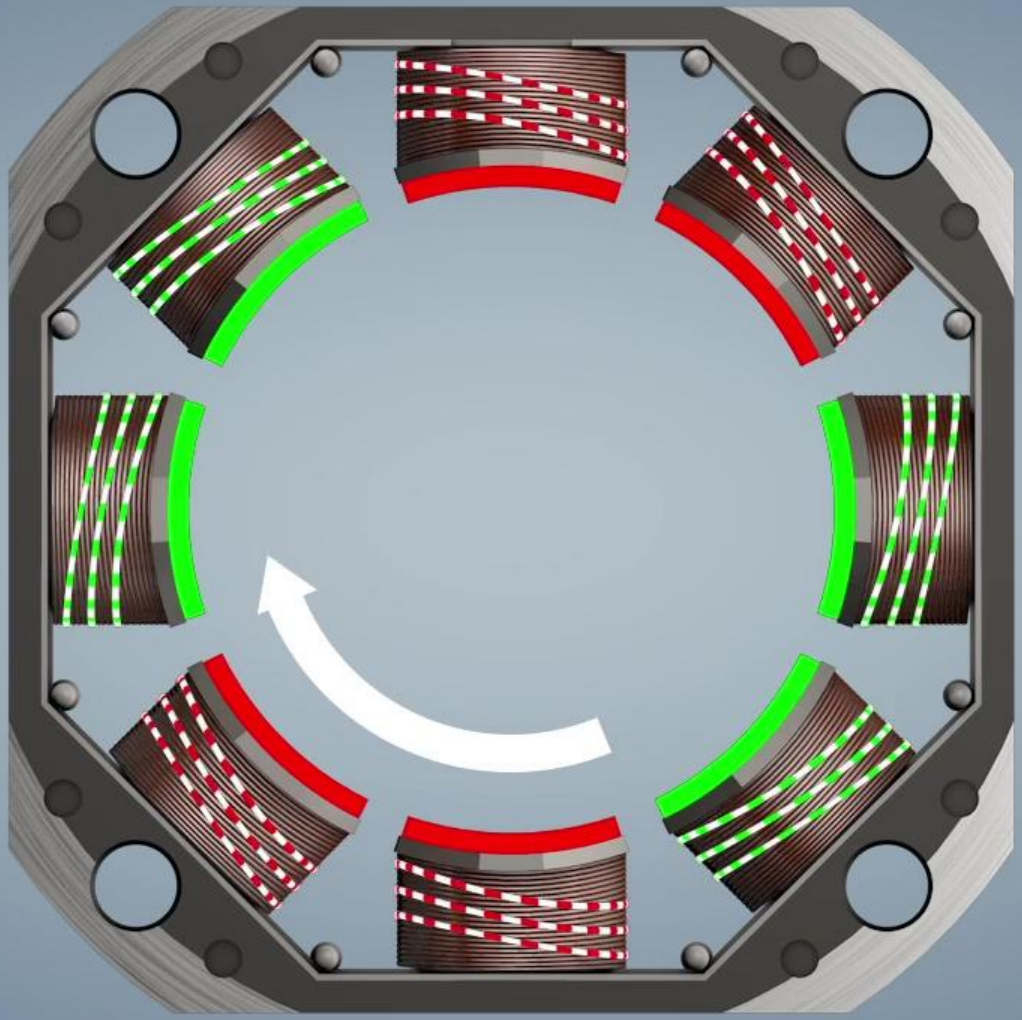


HOW STEPPER MOTOR WORKS



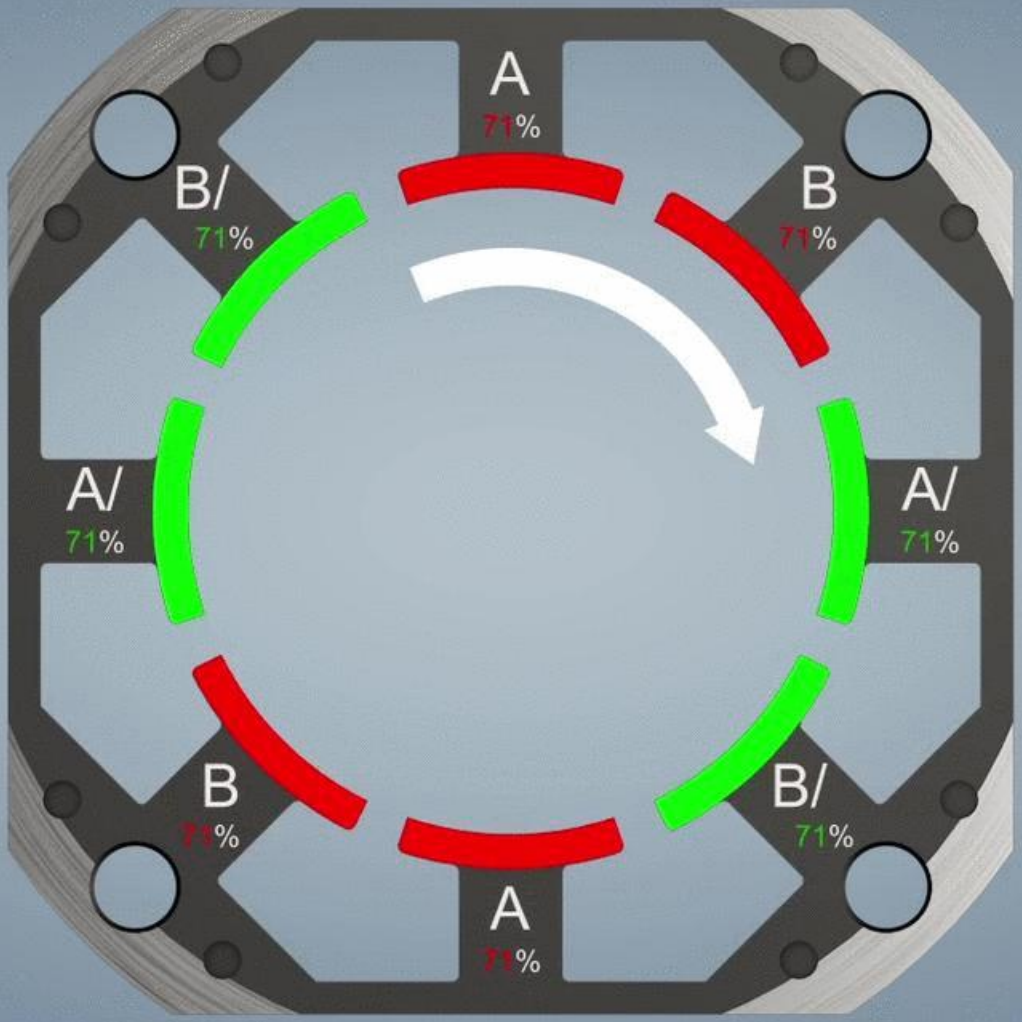
- Electrically, the stator part consists of two coils A and B
- The coils are derived in a certain sequence to produce rotating magnetic field

HOW STEPPER MOTOR WORKS

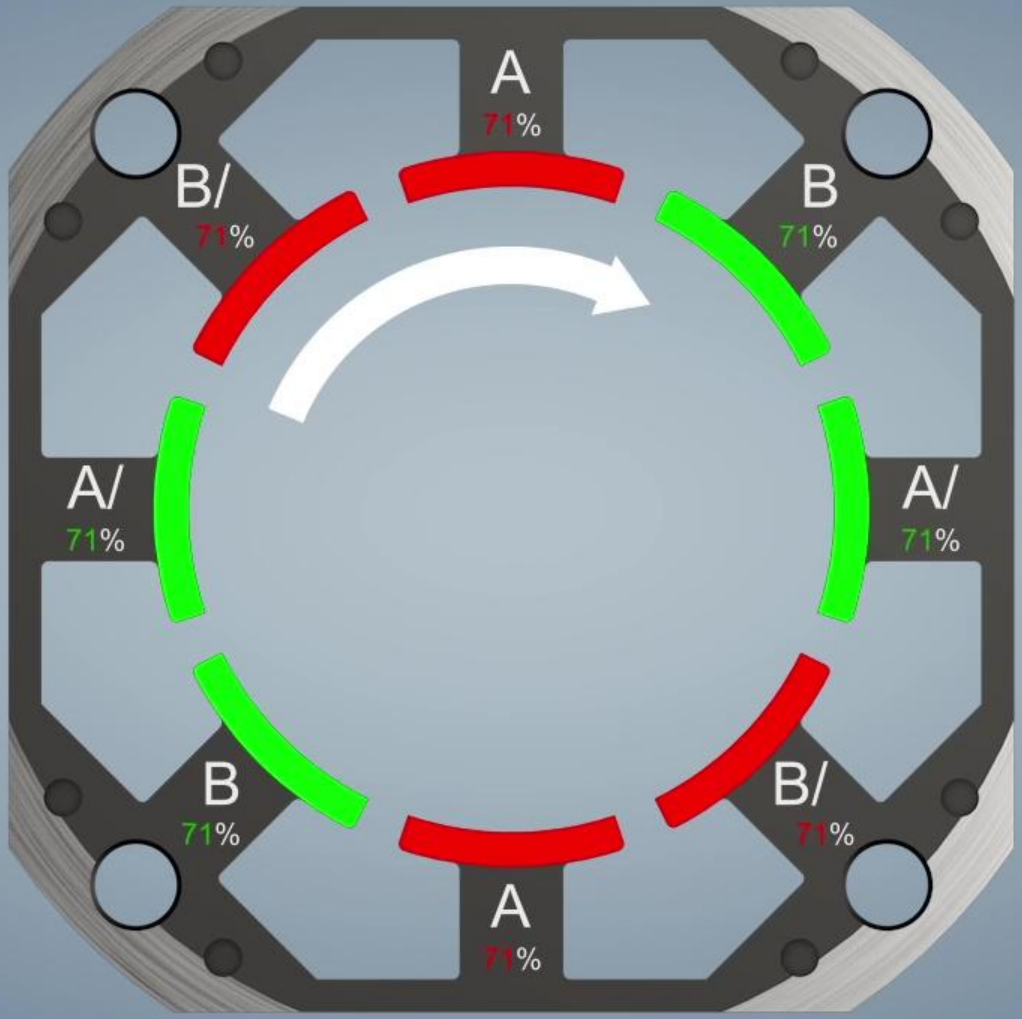


https://www.youtube.com/watch?v=Ew6eVGnj7r0

ROTATING MAGNETIC FIELD

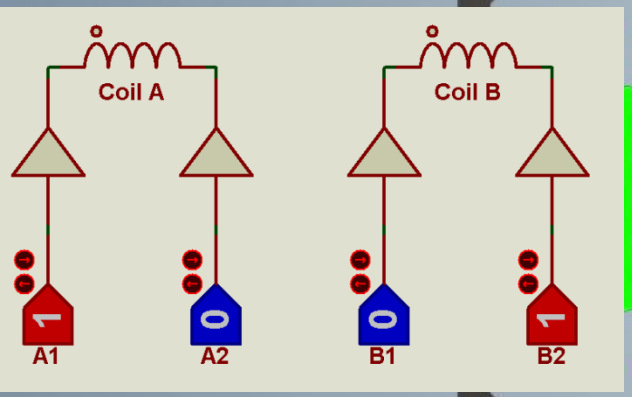


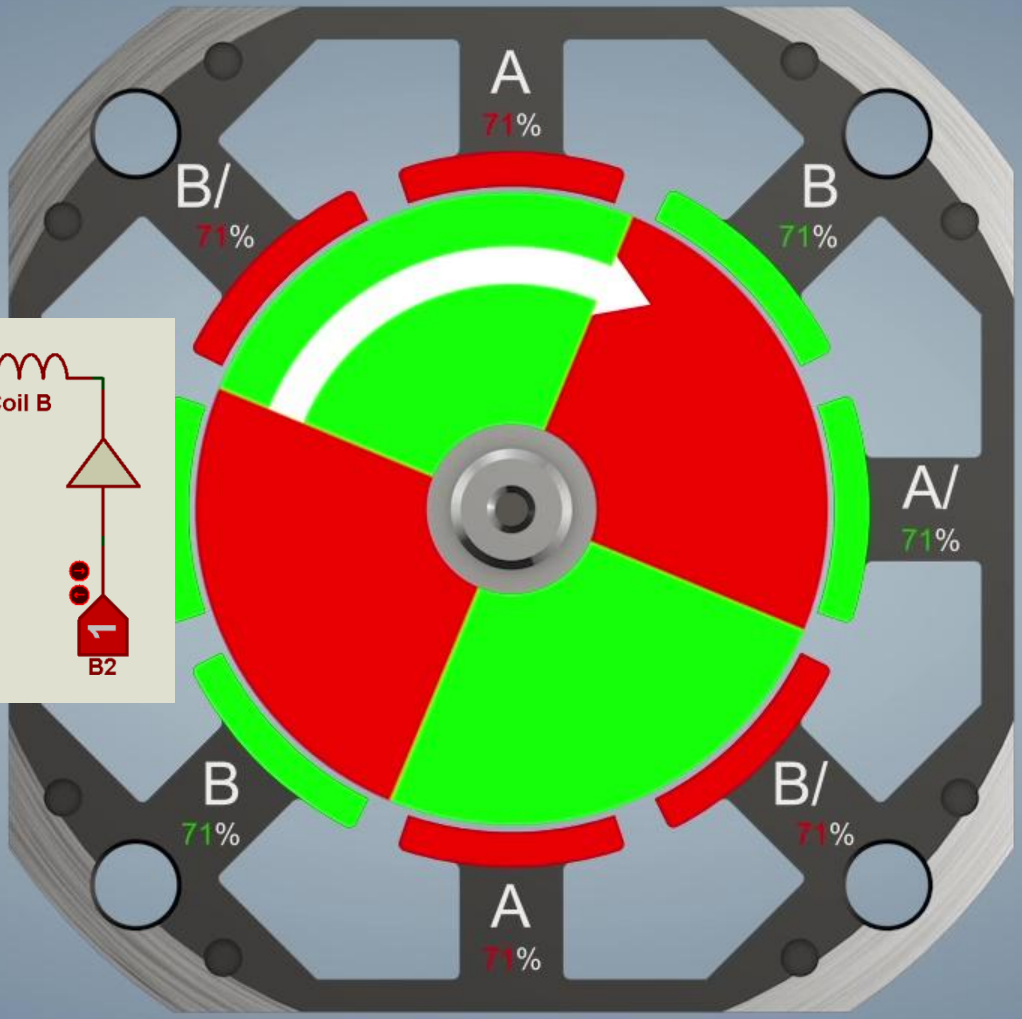
FULL STEP TWO PHASE MODE



- This mode energizes two coils at the same time to produce rotating magnetic field
- This type of control requires larger power but provides higher torque

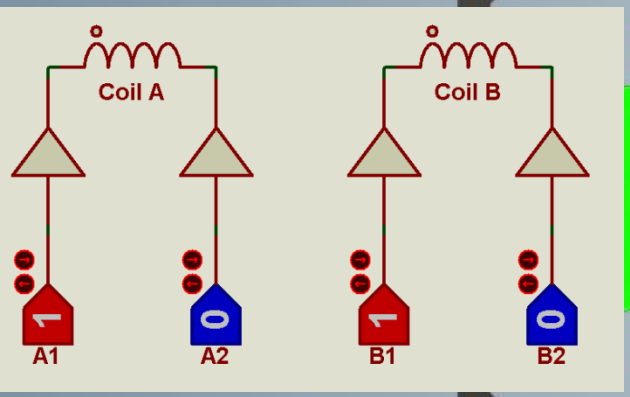
FULL STEP TWO PHASE MODE

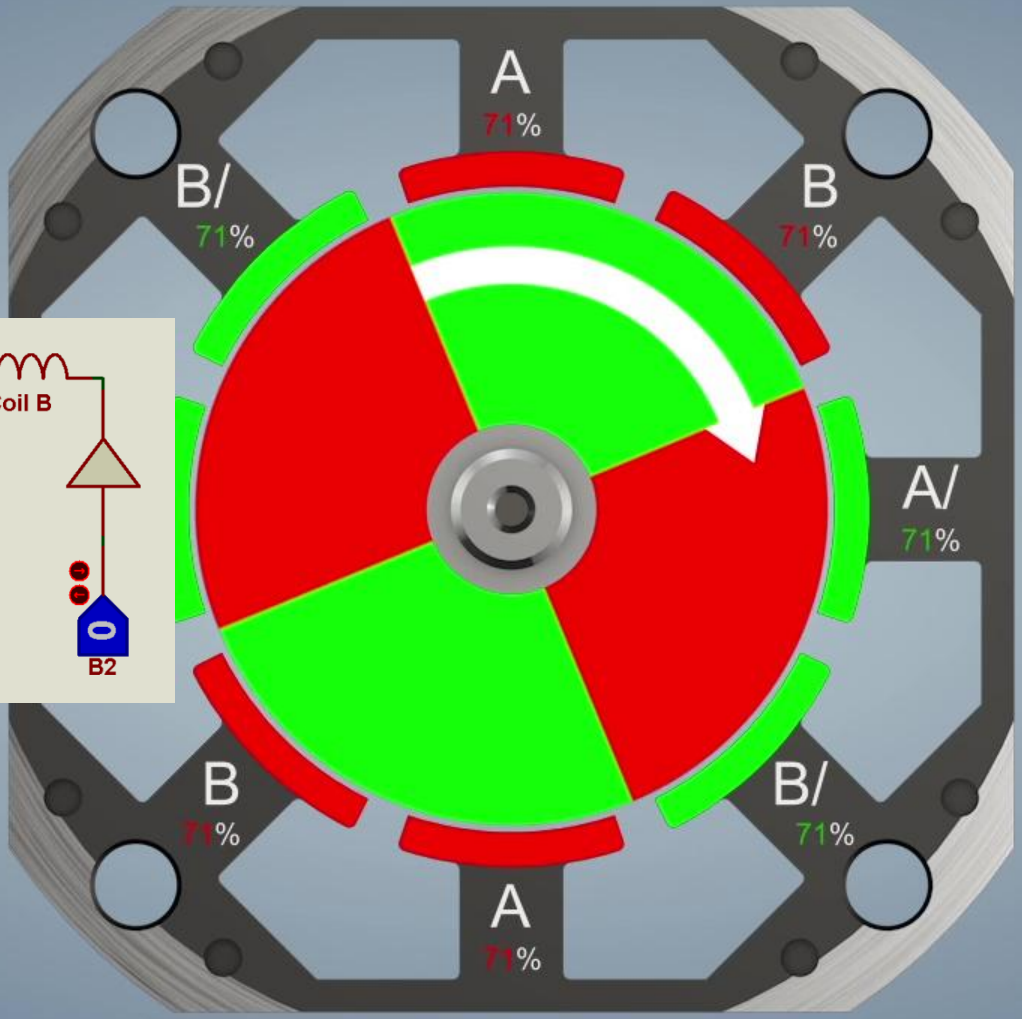




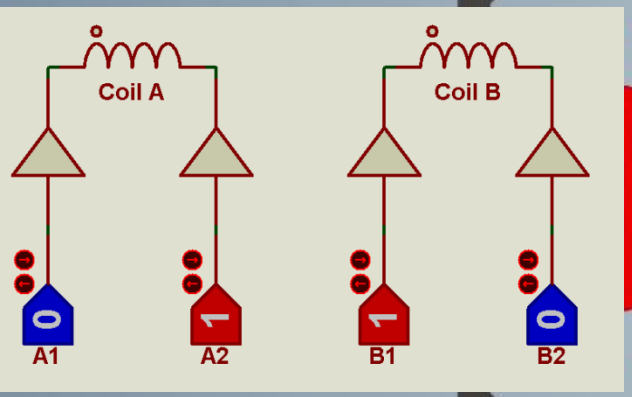
- The rotator is formed from a permanent magnet as shown
- The rotating magnetic field produced by the stator causes the rotator part to rotate

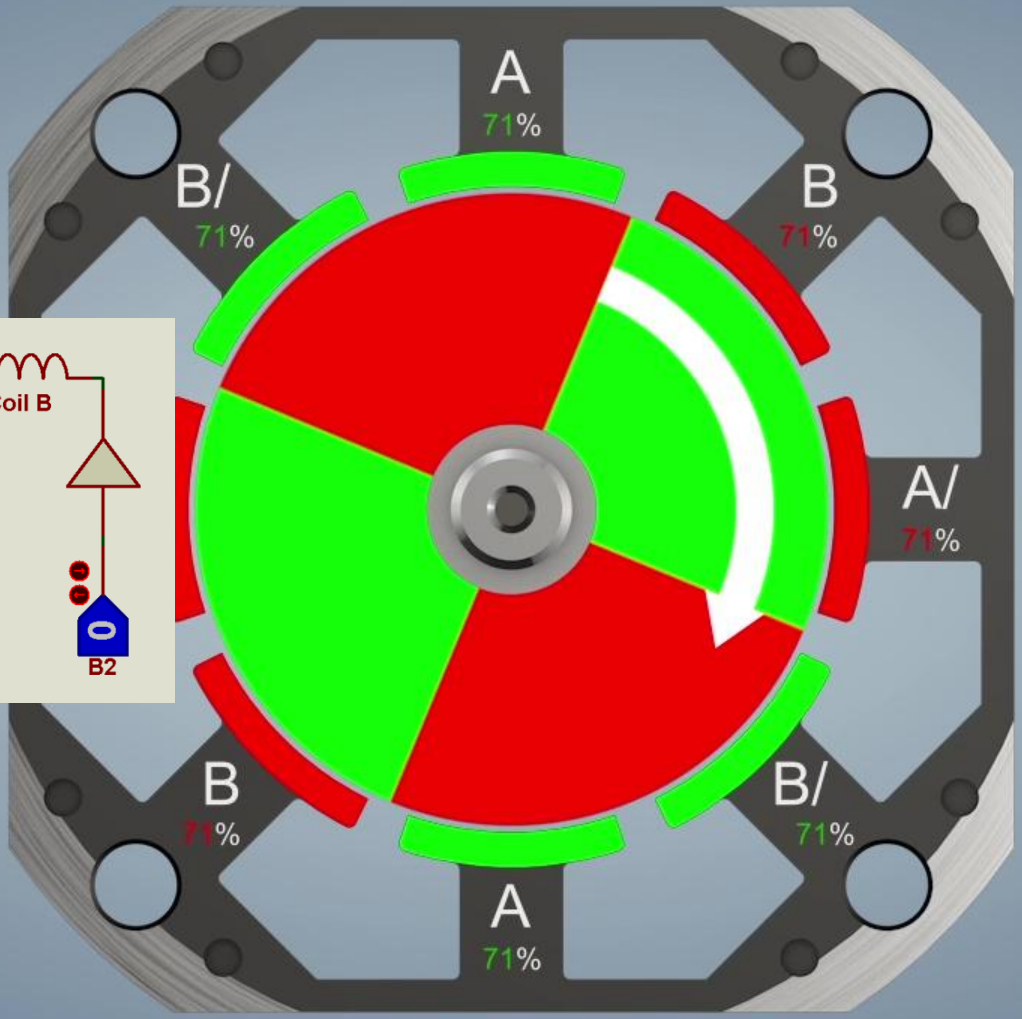
FULL STEP TWO PHASE MODE



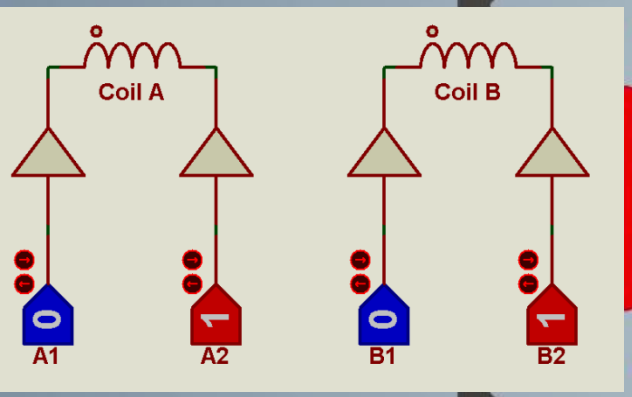


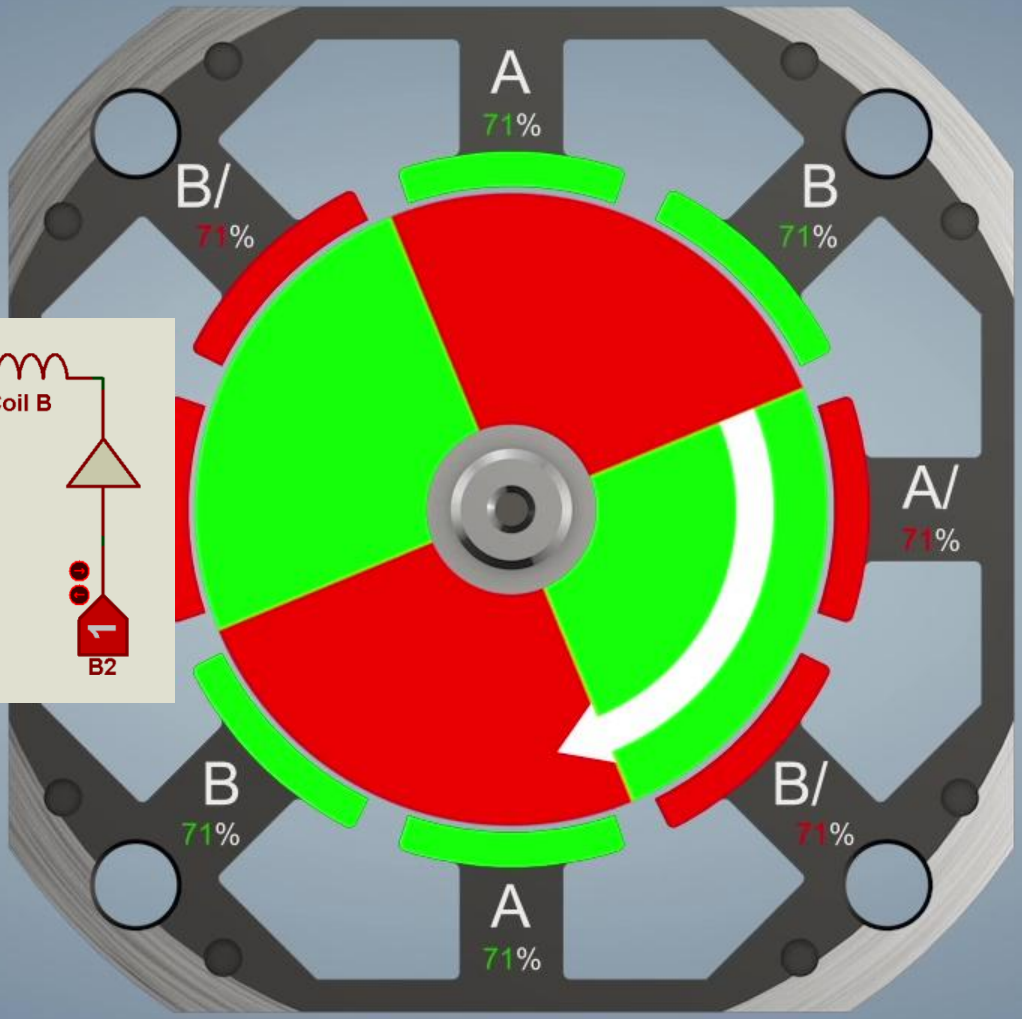
FULL STEP TWO PHASE MODE





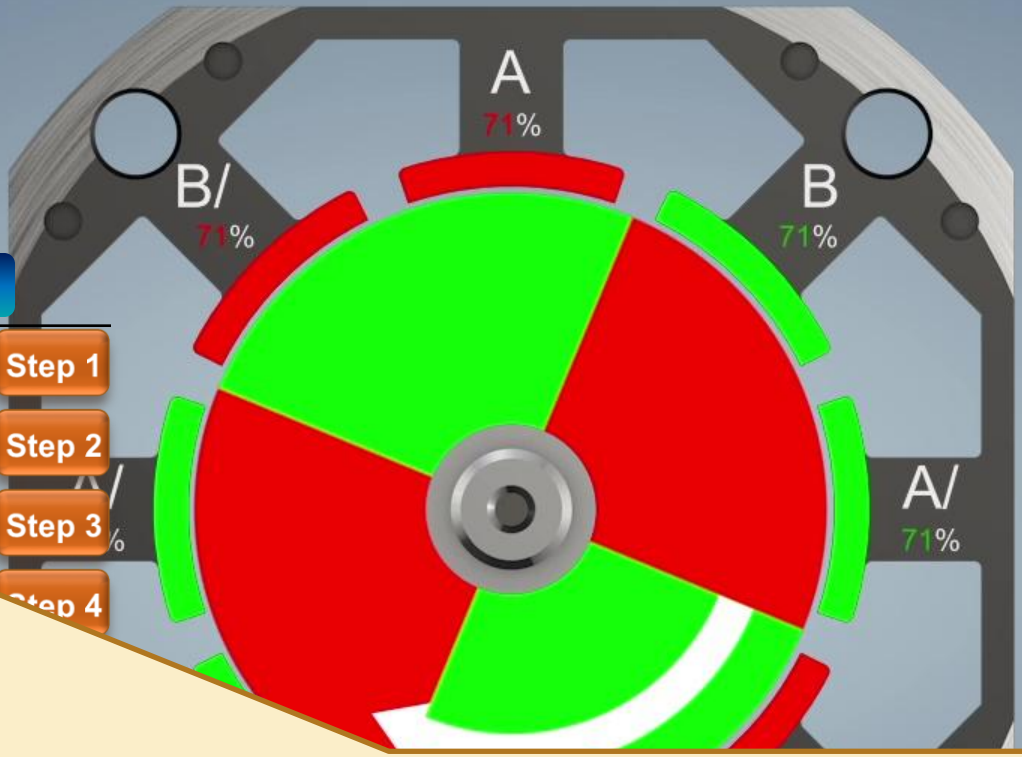
FULL STEP TWO PHASE MODE





FULL STEP TWO PHASE MODE

HEX	A1	A2	B1	B2	
9	1	0	0	1	Step 1
A	1	0	1	0	Step 2
6	0	1	1	0	Step 3
5	0	1	0		Step 4



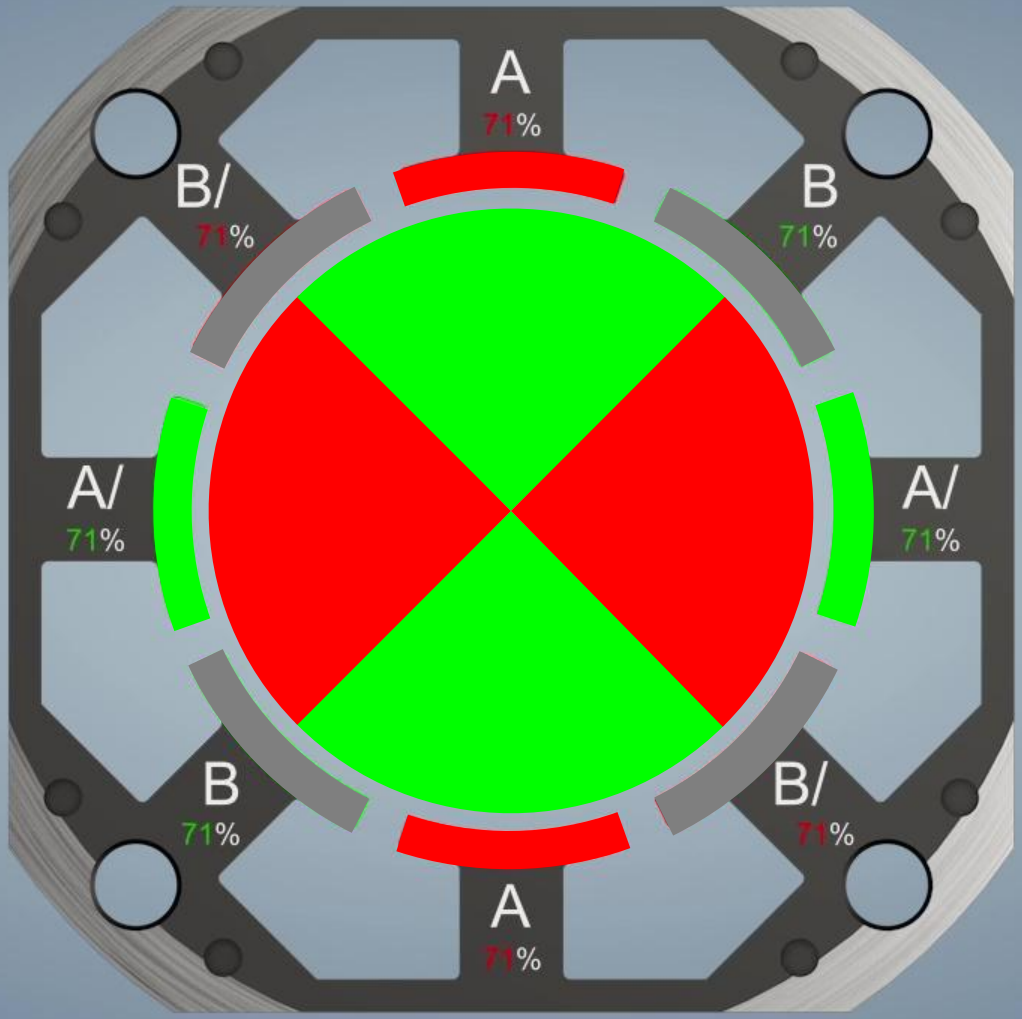
$$\alpha = \frac{360°}{2 \cdot m \cdot p} = \frac{360°}{2 \cdot 4 \cdot 2}$$

m = number of phases

p = number of pole pairs

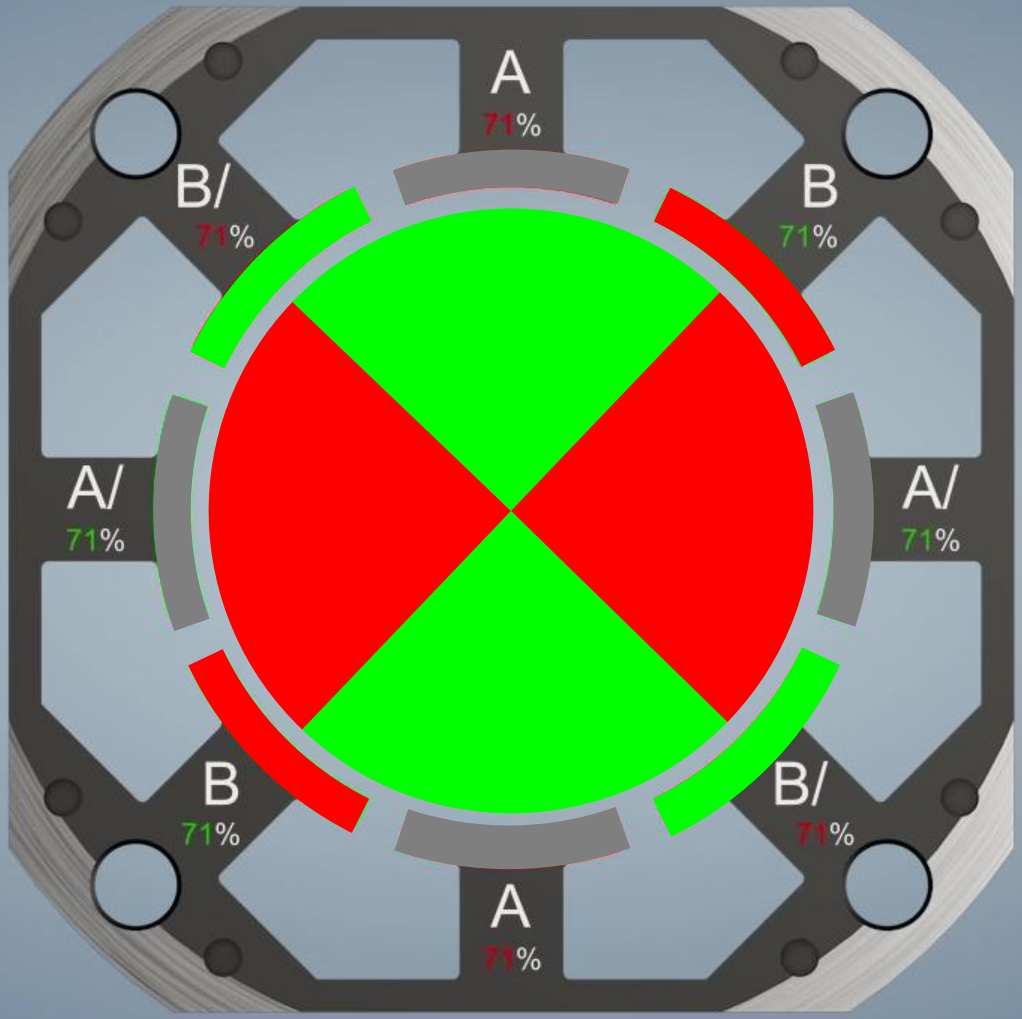
- To control stepper motor, we need to produce 4 digital sequence called steps
- In our example these sequence is 9, A, 6,5, and repeat
- Each step makes stepper motor to rotate by an angle of 45° clockwise
- To rotate counter-clockwise; send the steps in reverse order (5, 6, A, 9)
- In this example; full cycle rotation needs 8 steps (9, A, 6, 5, 9, A, 6, 5)
- The rotation speed is determined by the delay between steps
- In this example; for a rotation speed of 2 cycle/sec → step delay = 1/16 sec

FULL STEP ONE PHASE

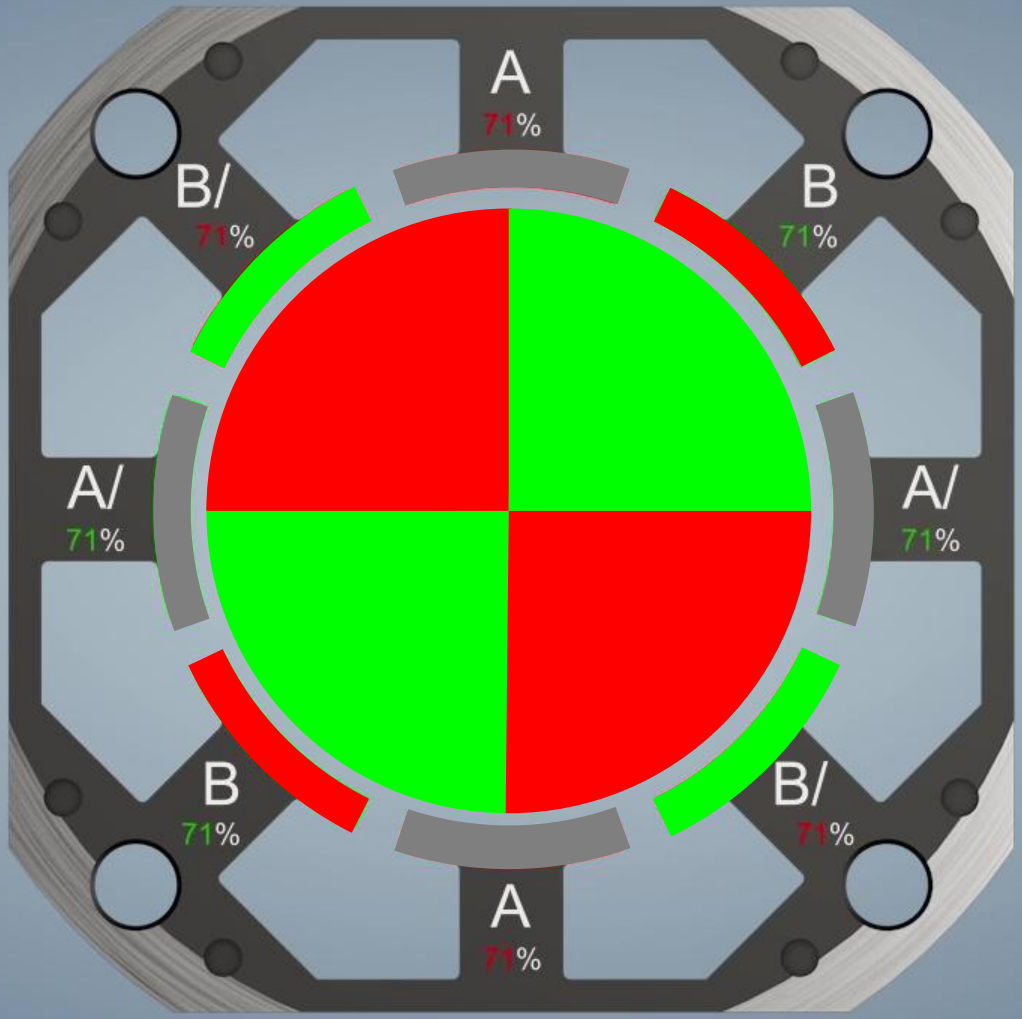


- This mode energizes only one coil at a time to produce rotating magnetic field
- This type of control requires the least amount of power but provides the least torque

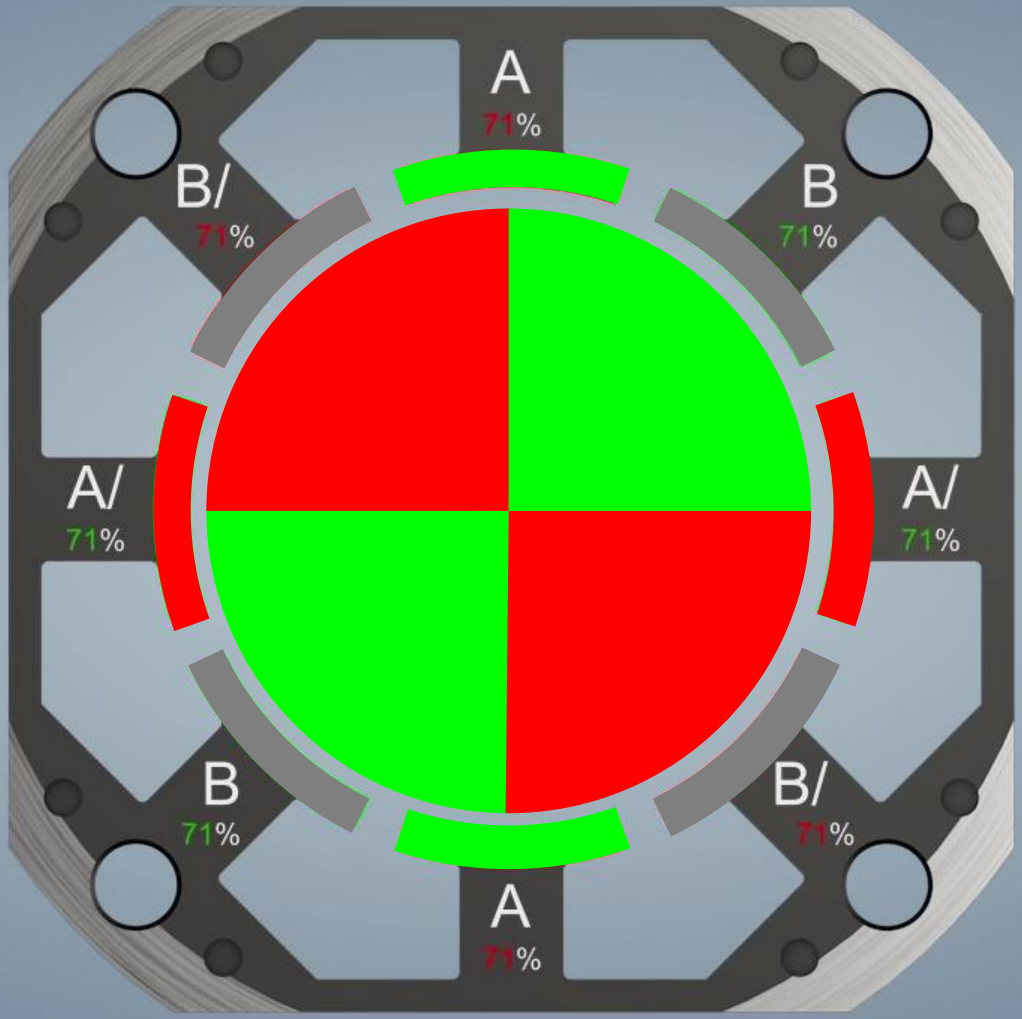
FULL STEP ONE PHASE MODE



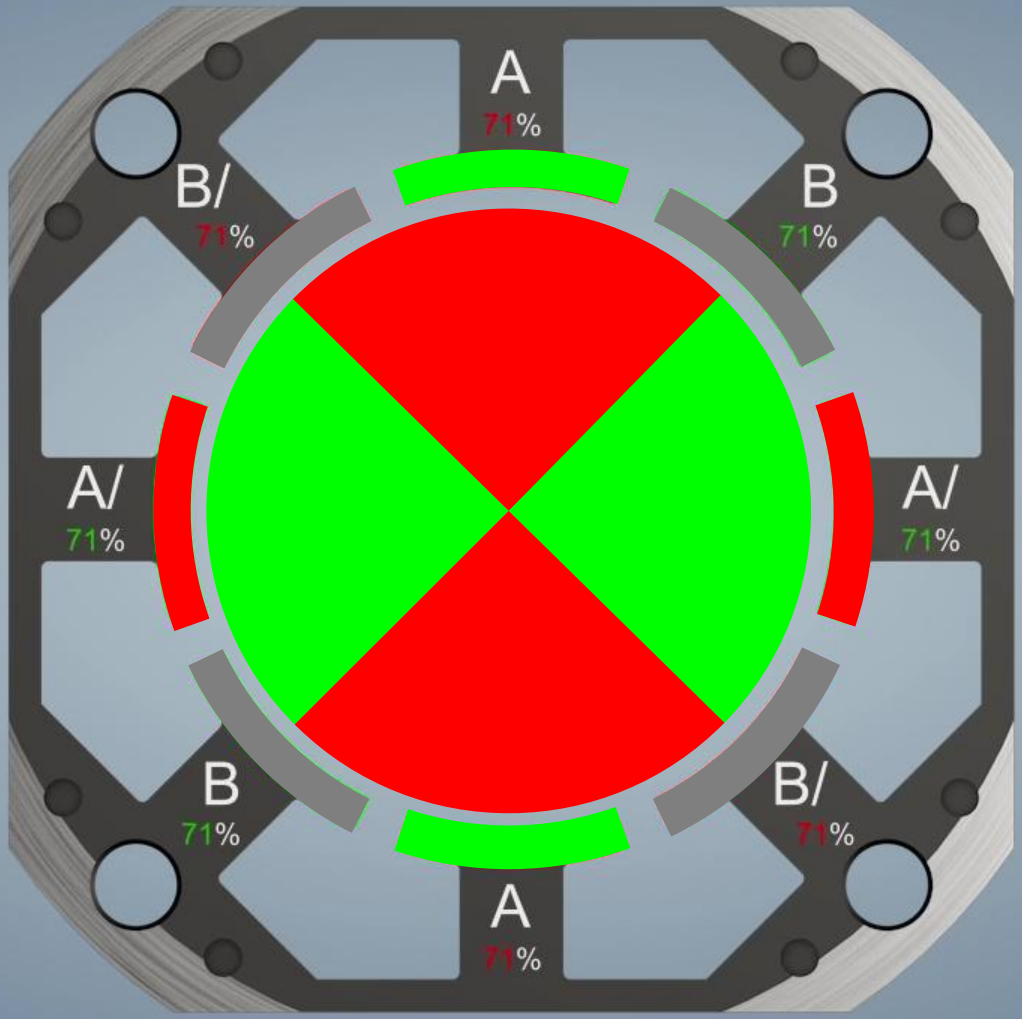
FULL STEP ONE PHASE MODE



FULL STEP ONE PHASE MODE

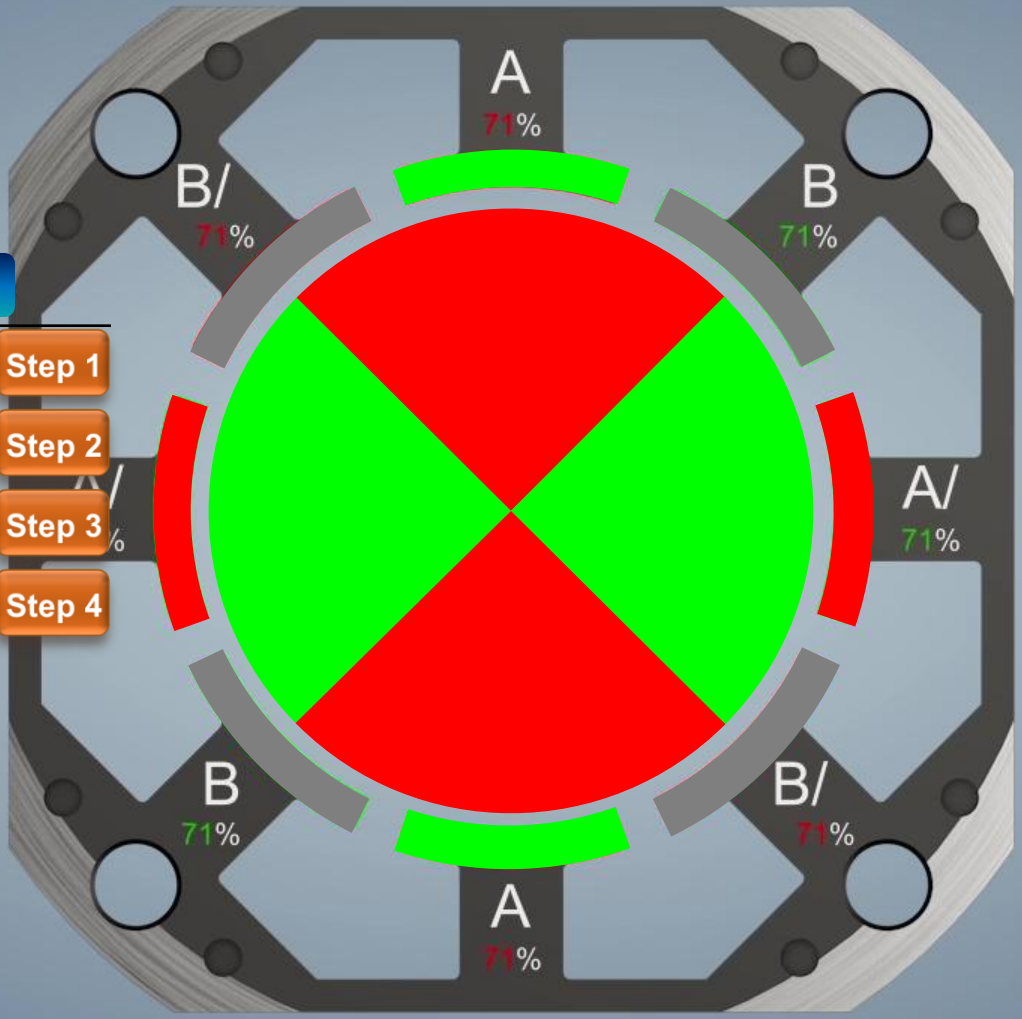


FULL STEP ONE PHASE MODE

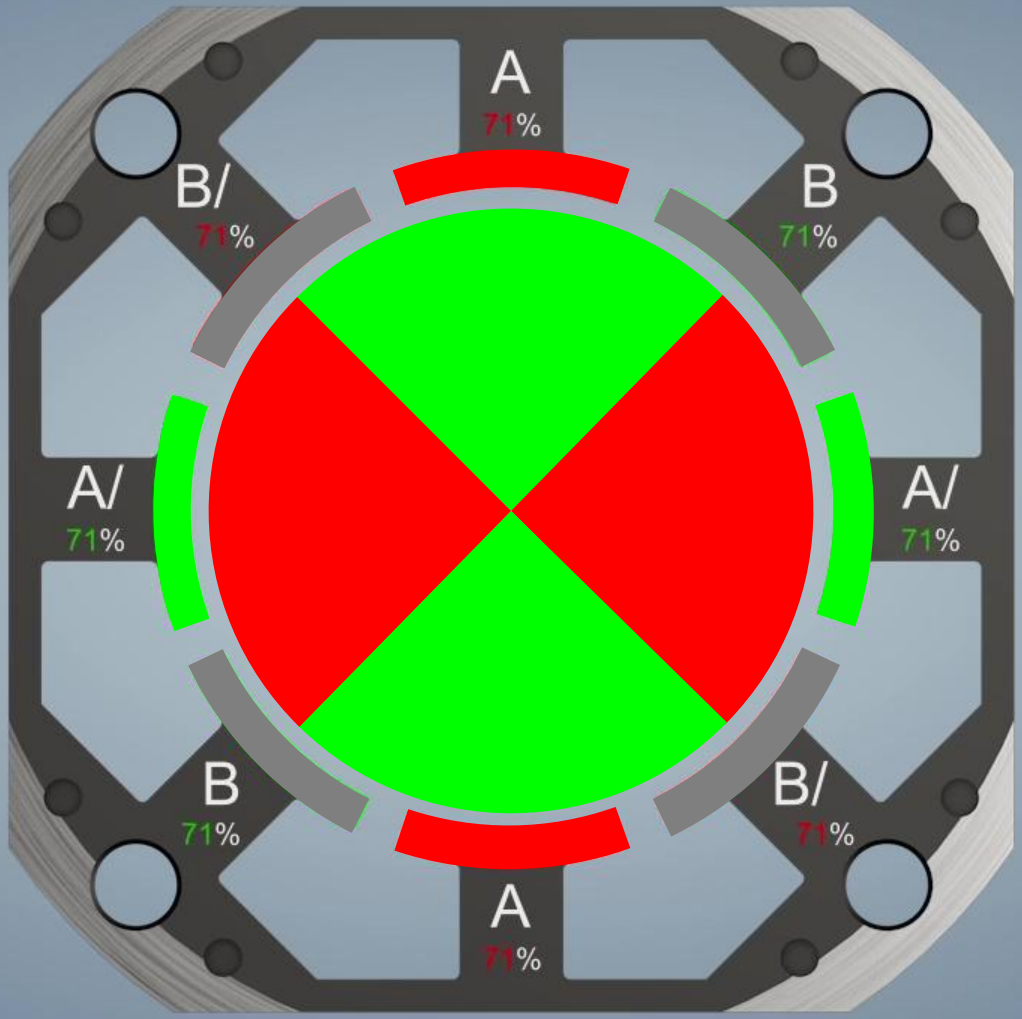


FULL STEP ONE PHASE MODE

HEX	A1	A2	B1	B2	
8	1	0	0	0	Step 1
2	0	0	1	0	Step 2
4	0	1	0	0	Step 3
1	0	0	0	1	Step 4

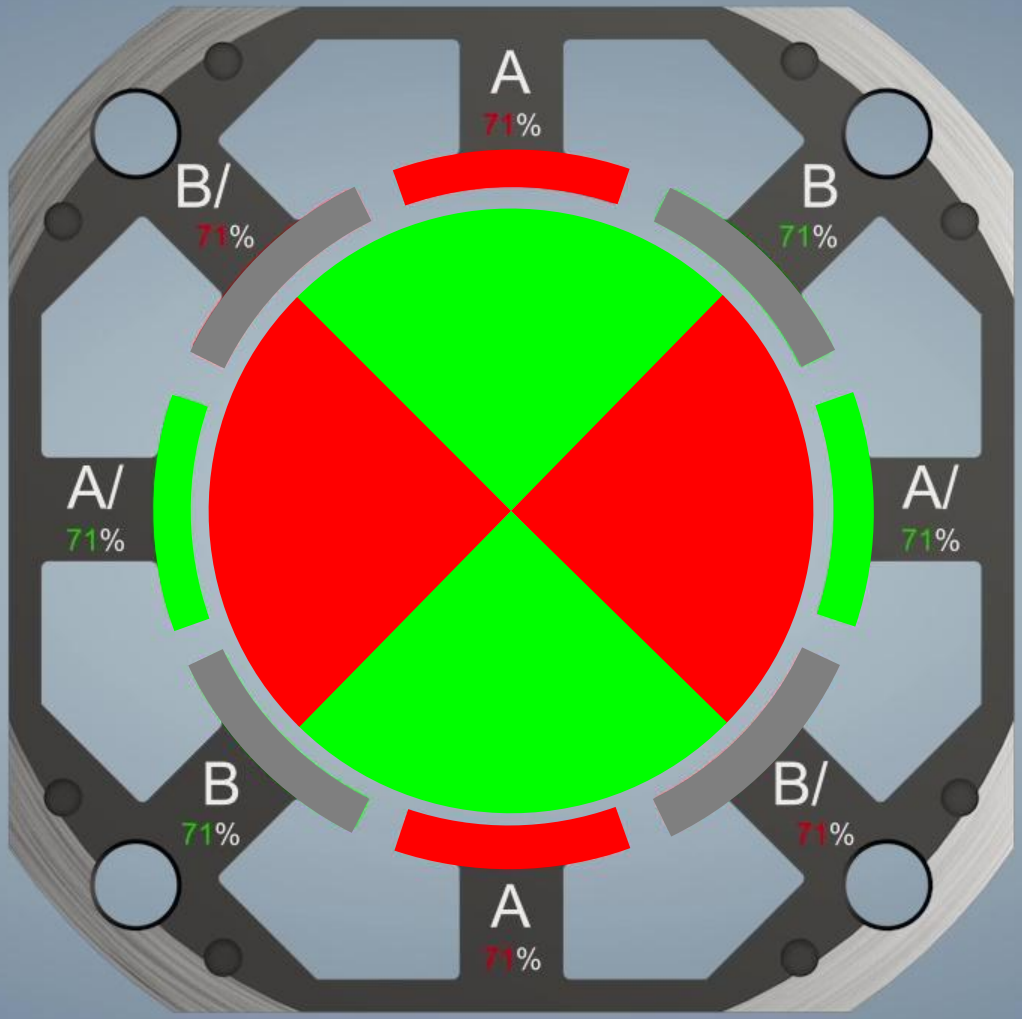


HALF STEP MODE

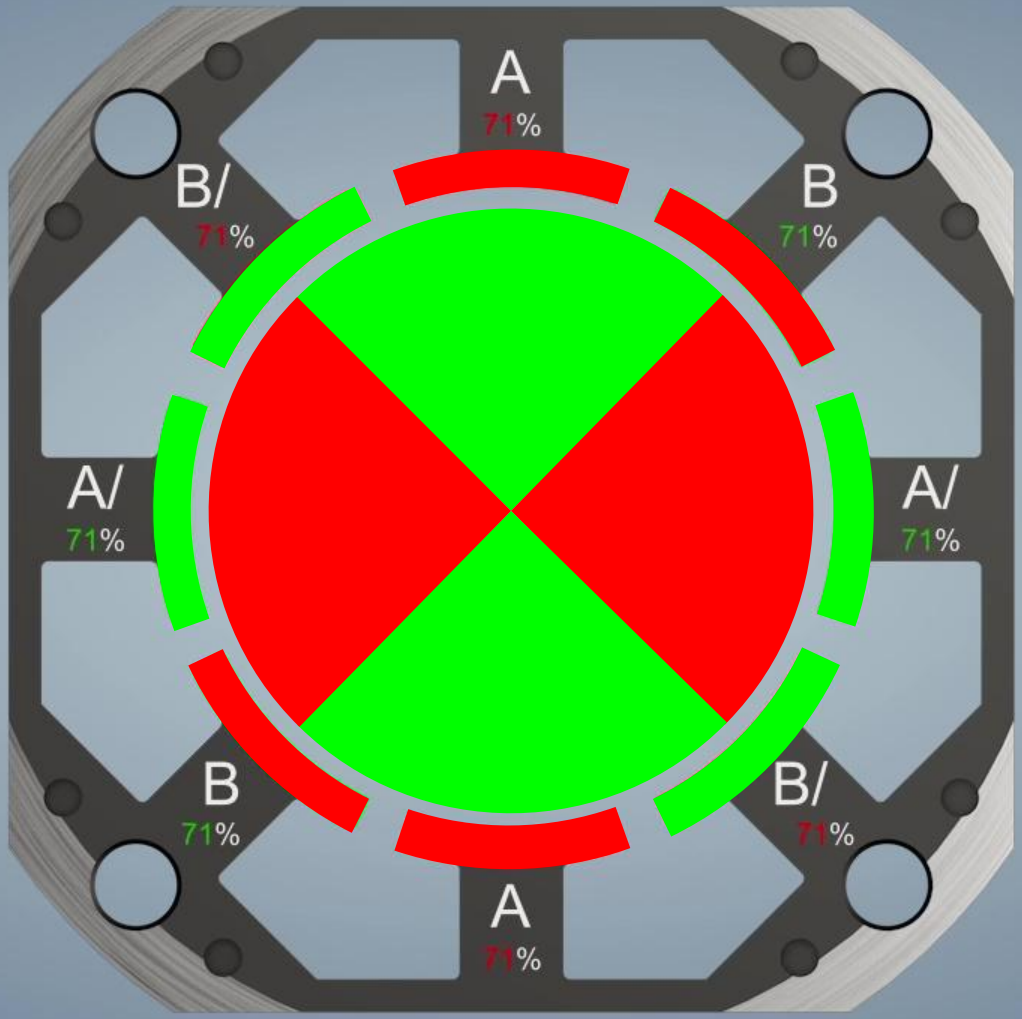


- Half step mode is a combination of the two full step modes (one and two-phase)
- The step angle is half that of previous modes (22.5 degree)

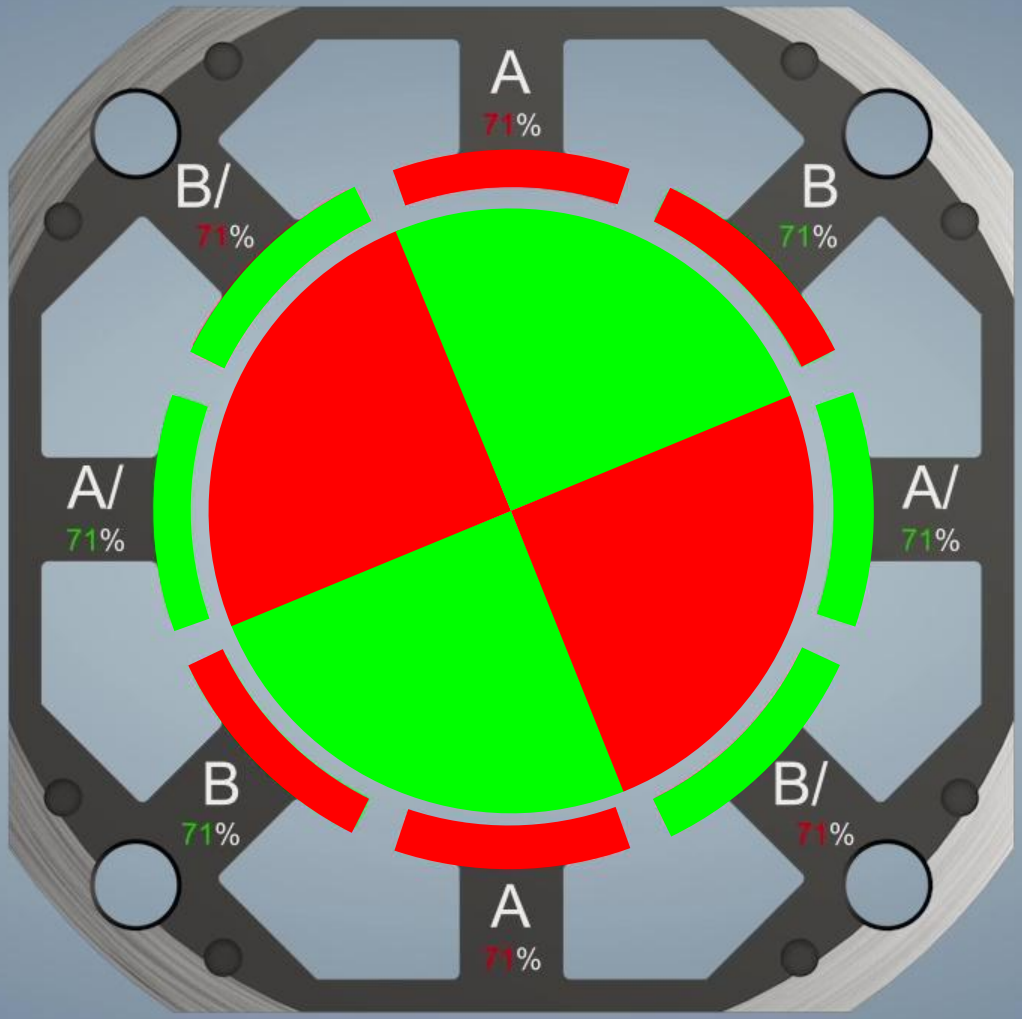
HALF STEP MODE



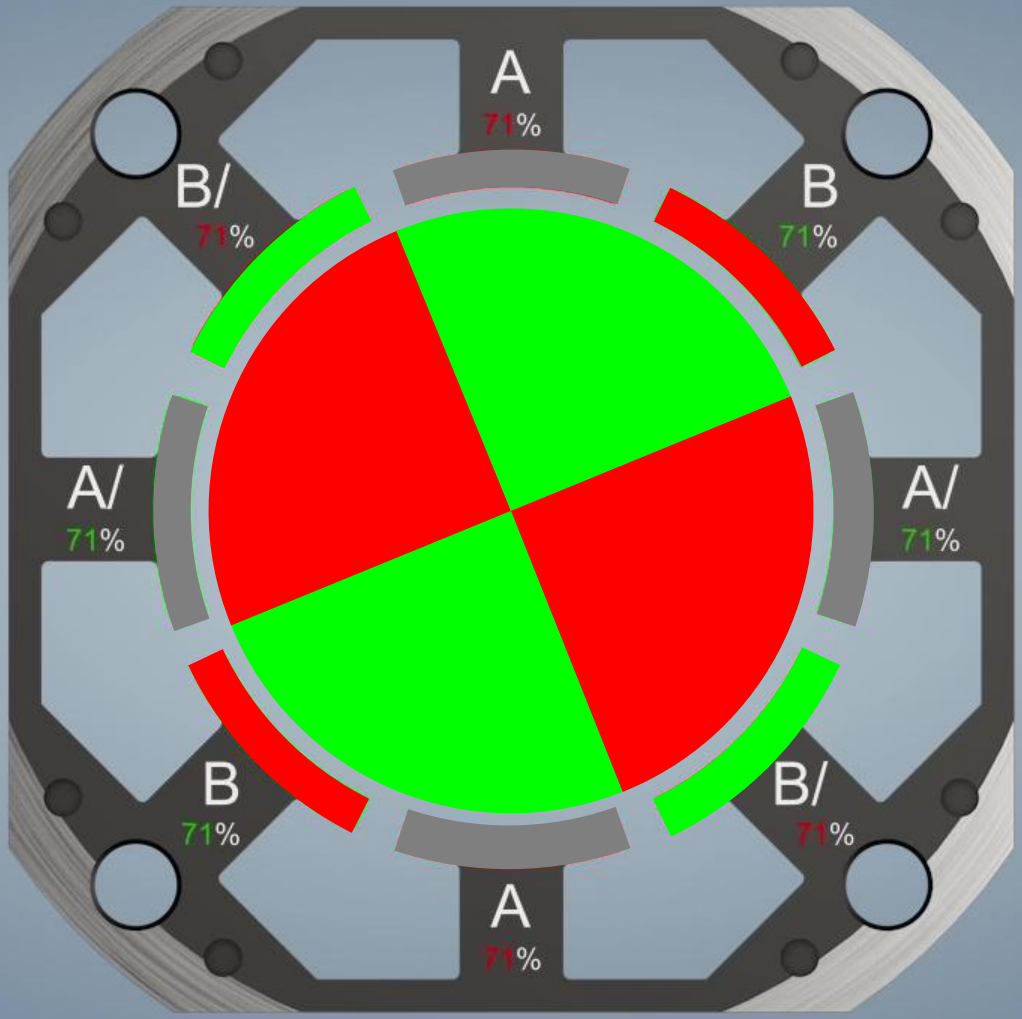
HALF STEP MODE



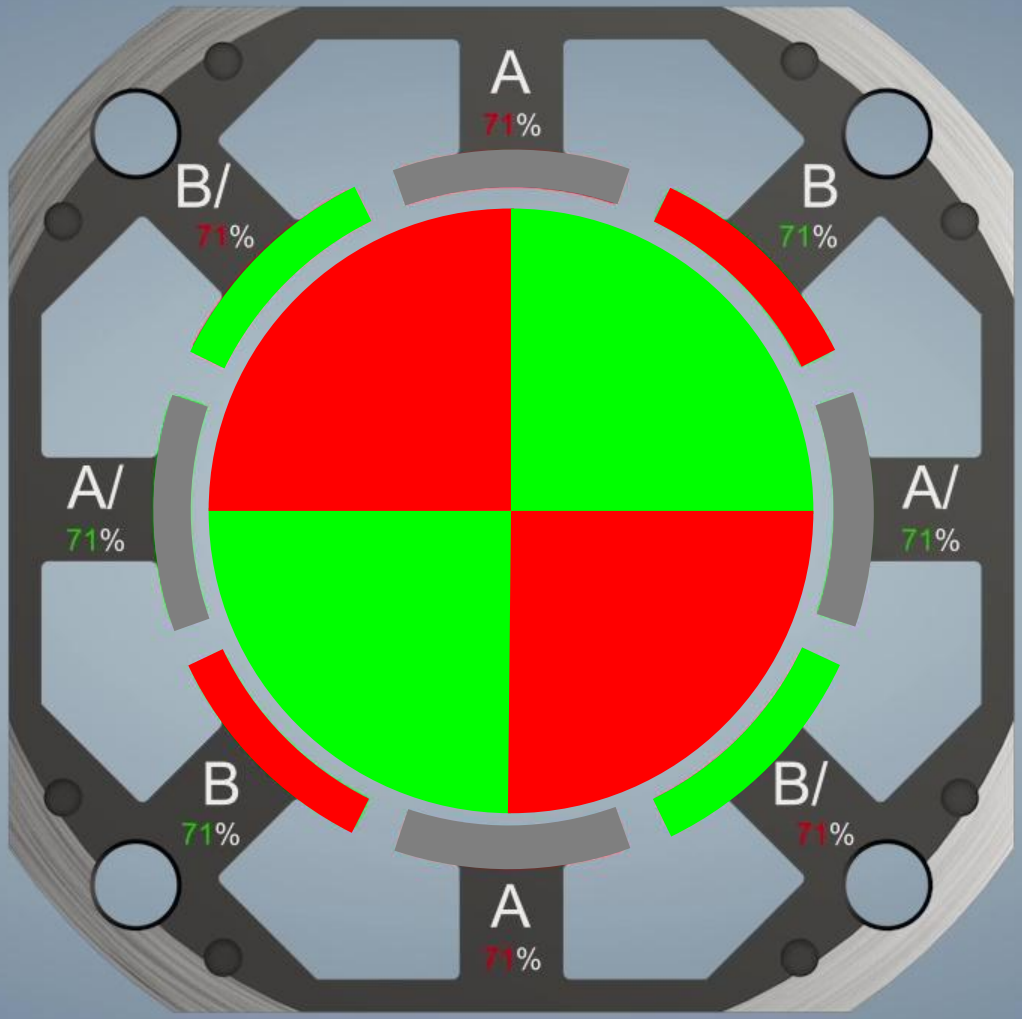
HALF STEP MODE



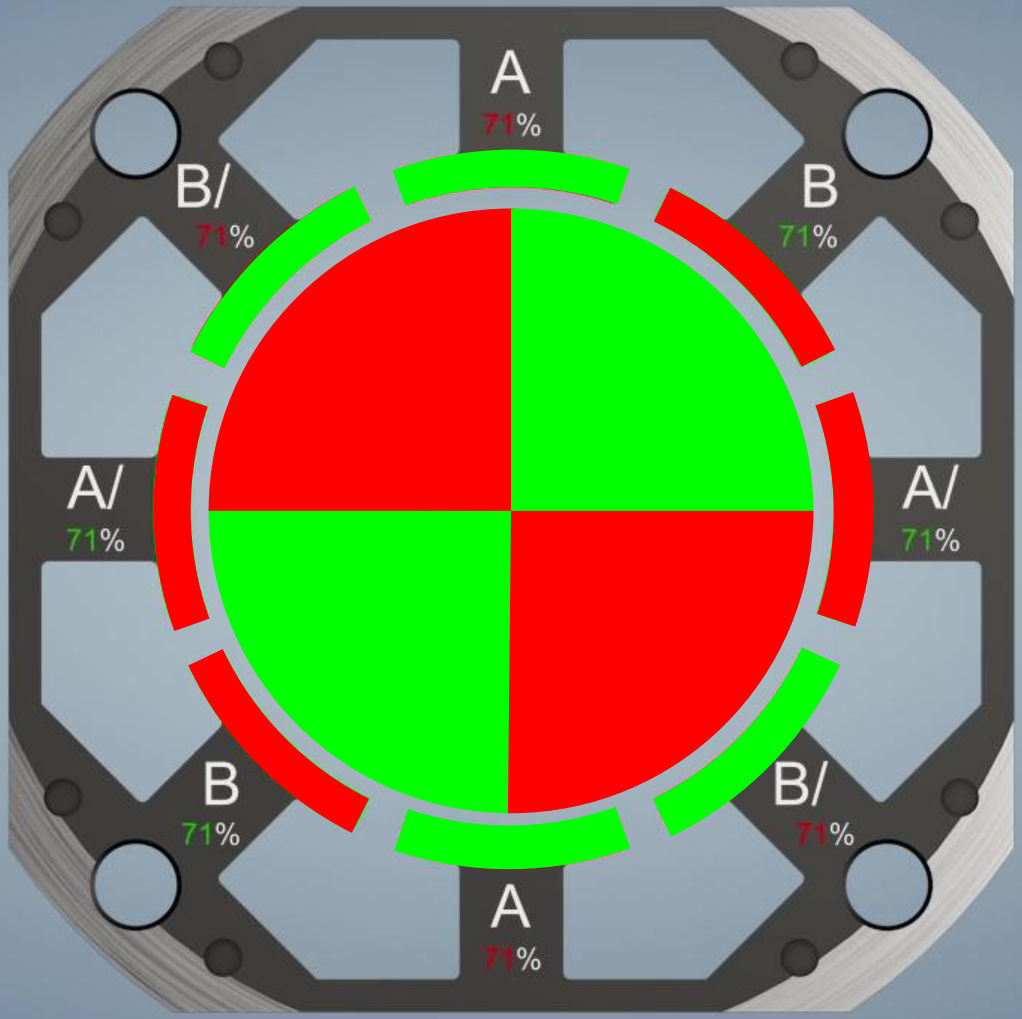
HALF STEP MODE



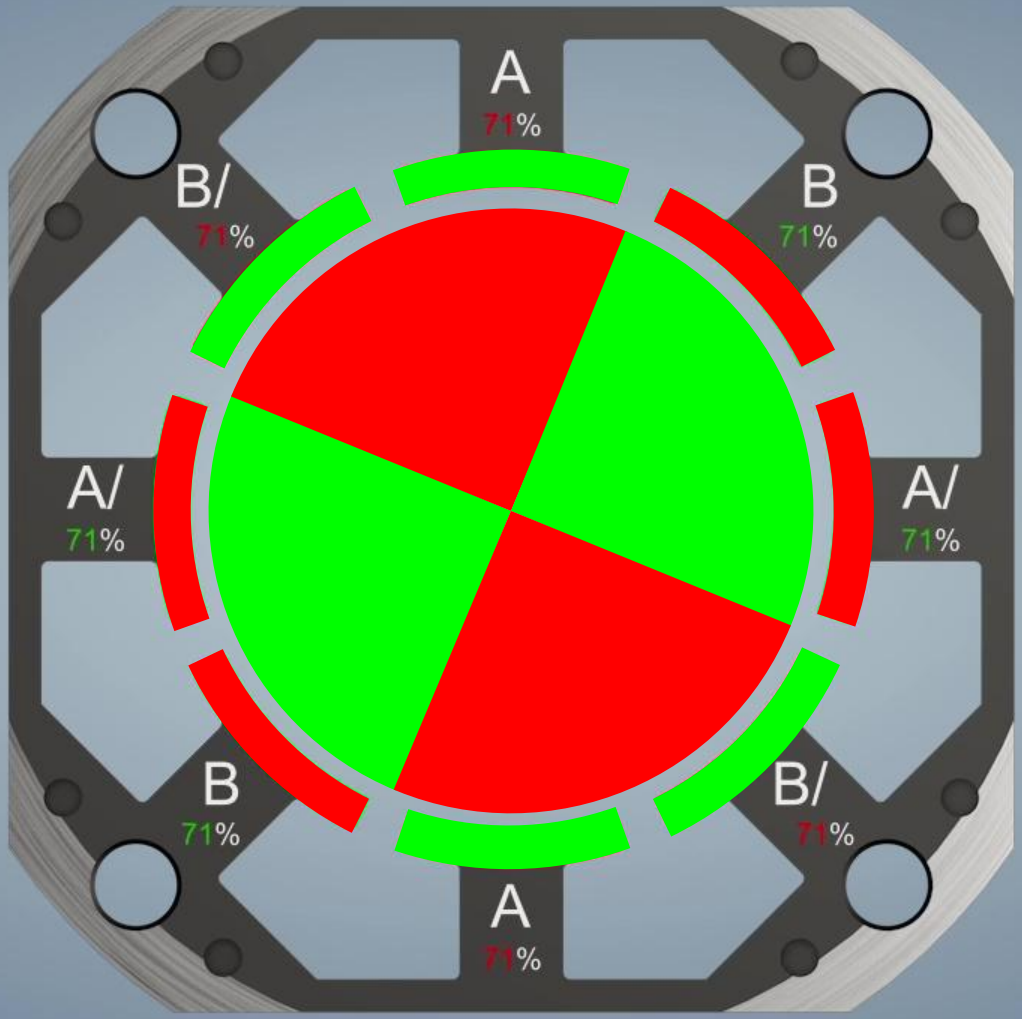
HALF STEP MODE



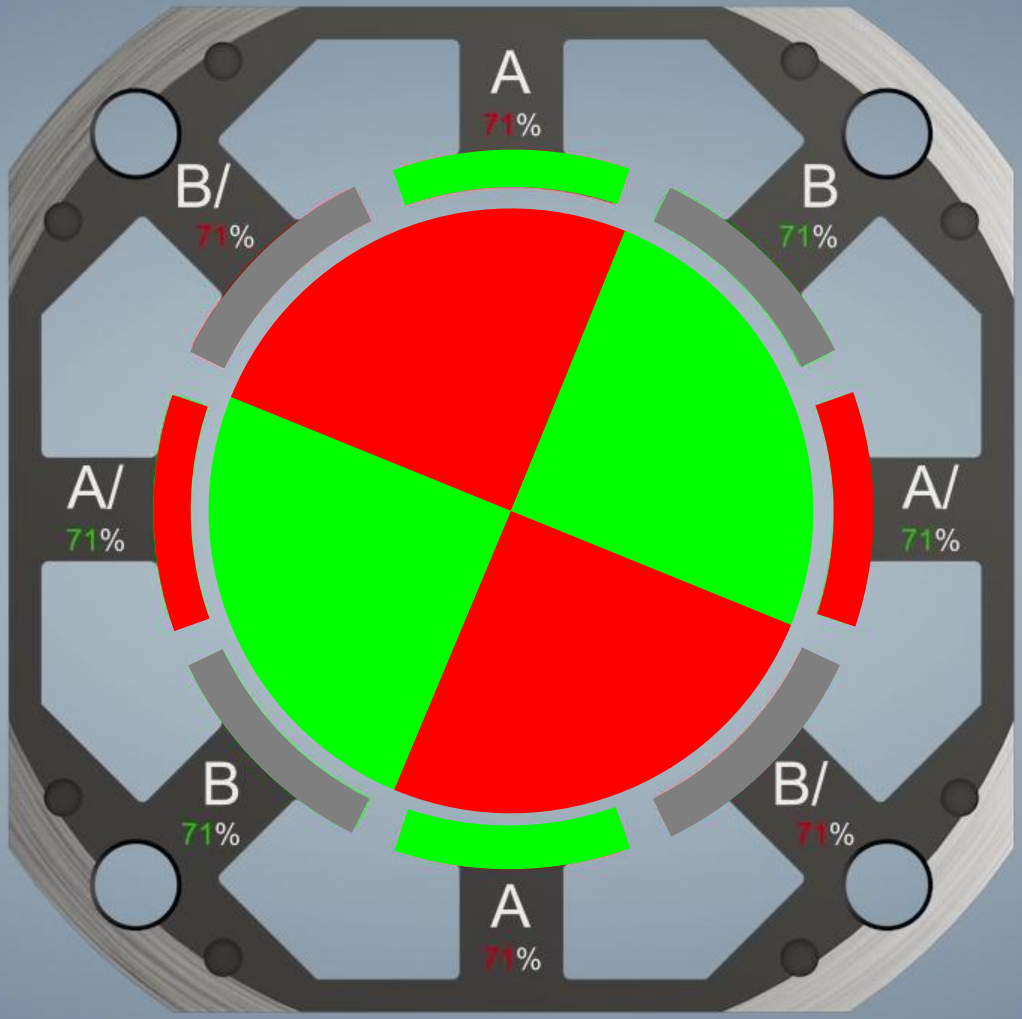
HALF STEP MODE



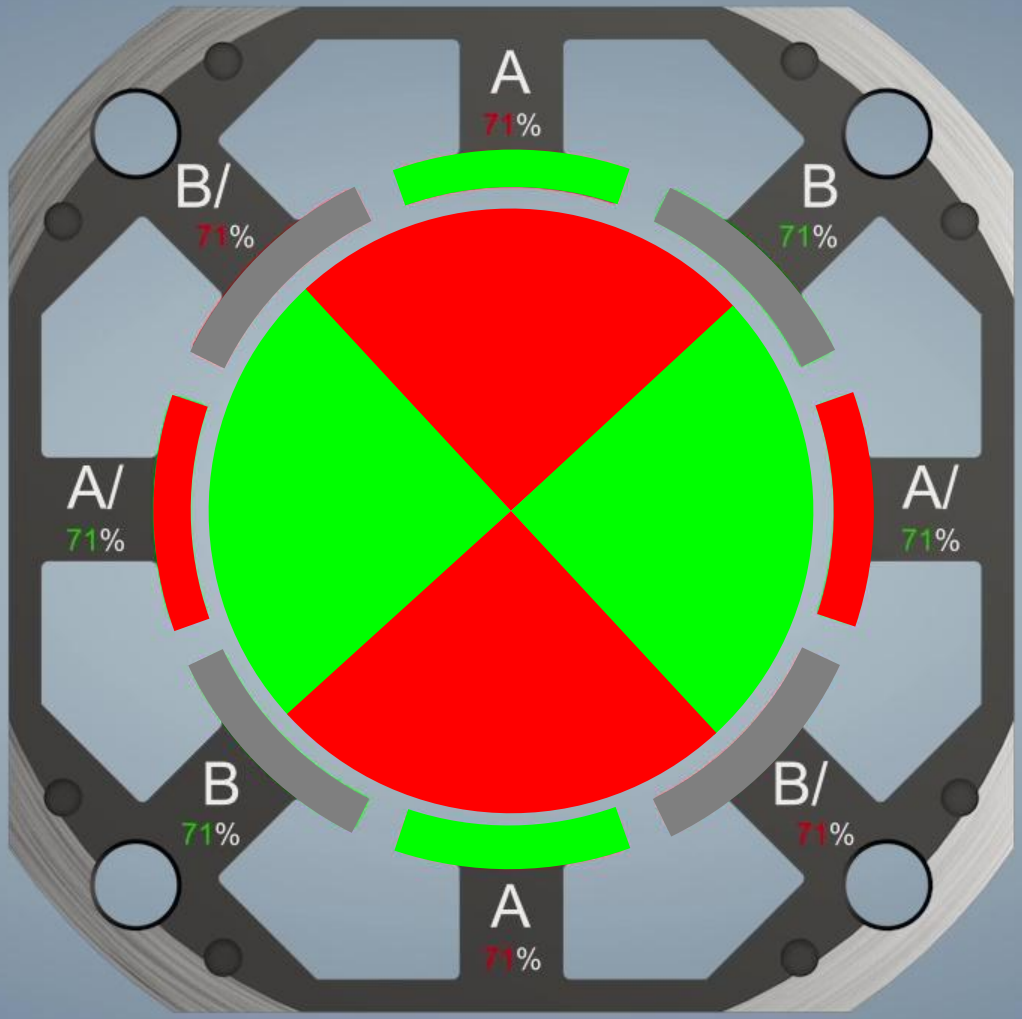
HALF STEP MODE



HALF STEP MODE

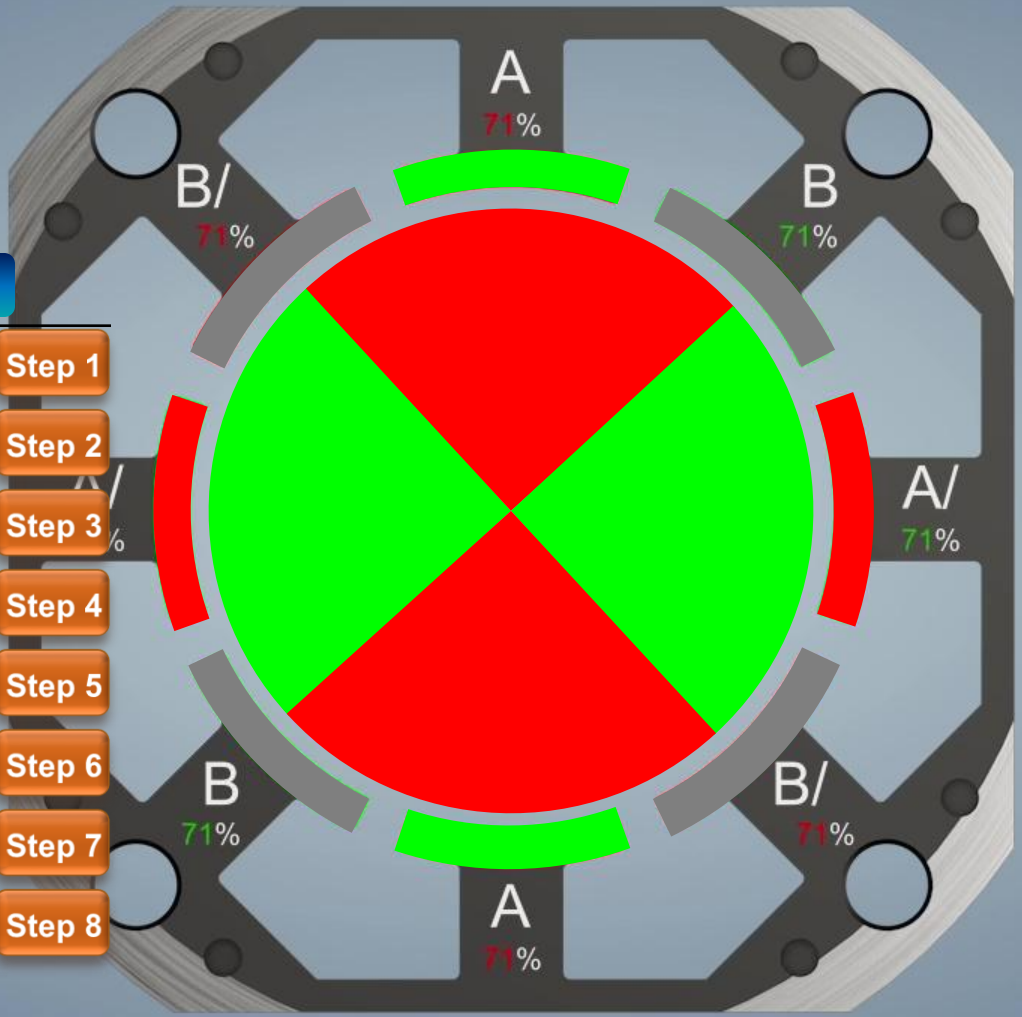


HALF STEP MODE



HALF STEP MODE

HEX	A1	A2	B1	B2	
8	1	0	0	0	Step 1
A	1	0	1	0	Step 2
2	0	0	1	0	Step 3
6	0	1	1	0	Step 4
4	0	1	0	0	Step 5
5	0	1	0	1	Step 6
1	0	0	0	1	Step 7
9	1	0	0	1	Step 8

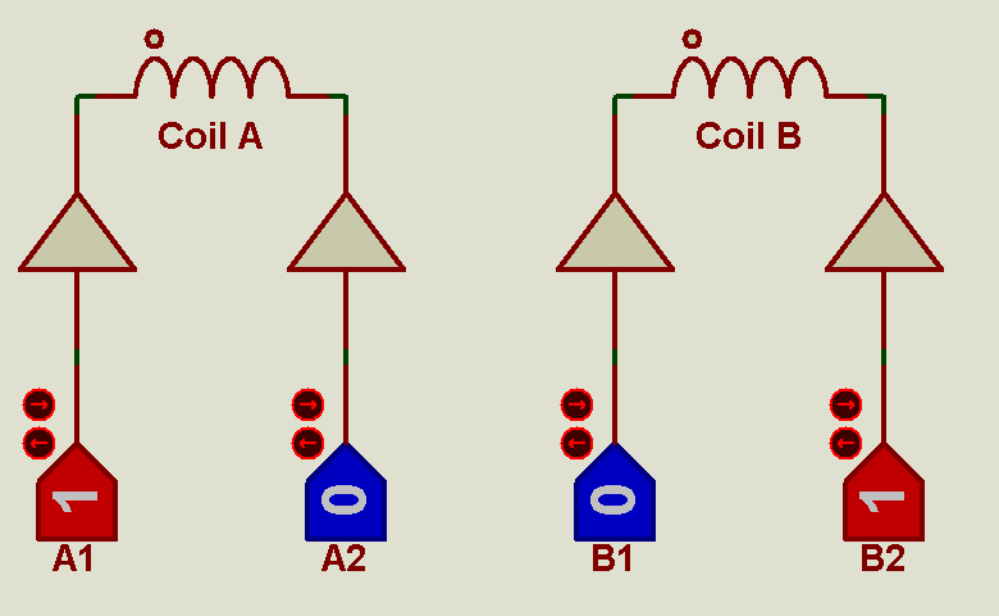


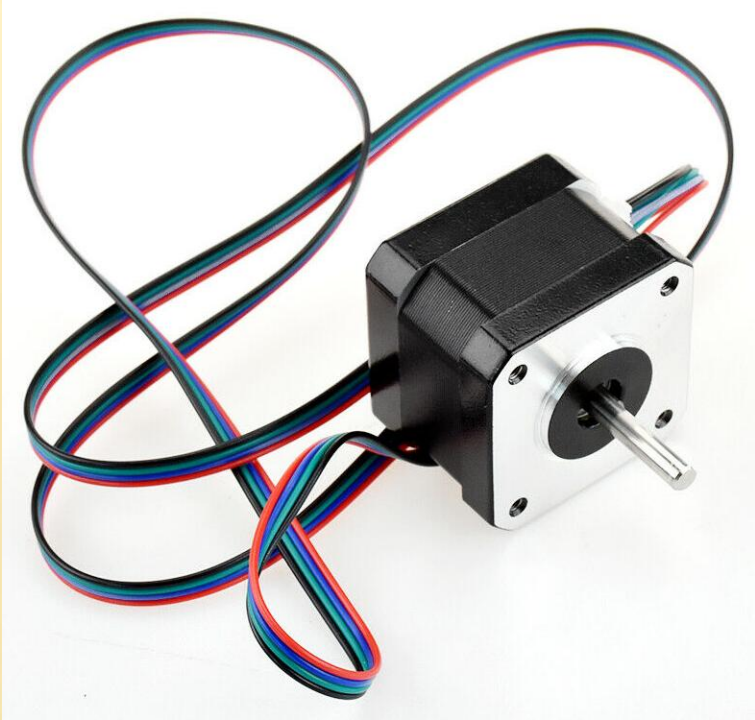
STEPPER MOTORS CHARACTERSTICS

- Move using control modules that energize the coils in the stator in a certain pattern that lines the magnetic poles in the rotor with the appropriate charge (N or S).
- When incremental rotary motion is required, it is possible to use **stepper motors**
- A stepper motor possesses the ability to move a specified number of revolutions or fraction of a revolution in order to achieve a fixed and consistent angular movement
- Stepper motors are another kind of motors that do not require feedback
- A stepper motor can be incrementally driven, one step at a time, forward or backward
- Stepper motor characteristics are:
  - Number of steps per revolution (e.g. 200 steps per revolution → 1.8° per step)
  - Max. number of steps per second ("stepping rate" → max speed)
- Driving a stepper motor requires a 4 step switching sequence for full-step mode
- Stepper motors can also be driven in 8 step switching sequence for half-step mode (higher resolution)



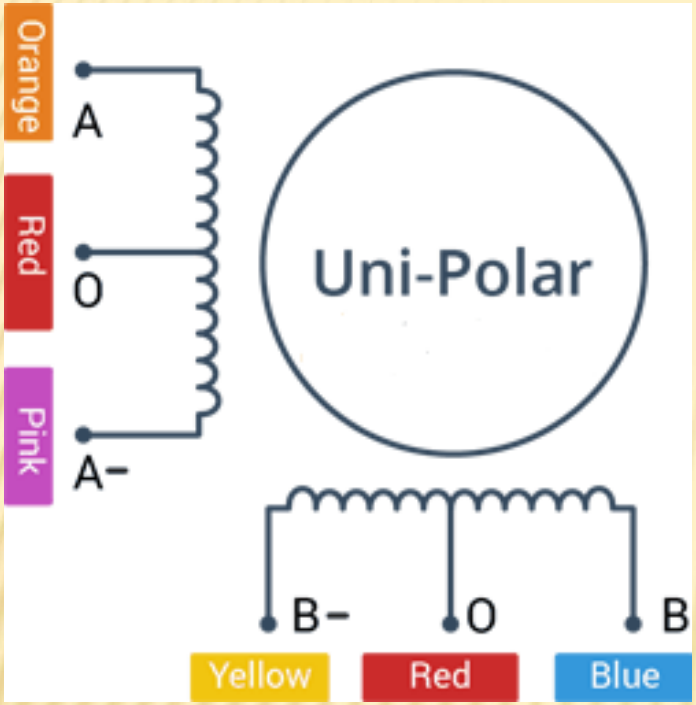
BIPOLAR STEPPER MOTORS





- Each pole consists of only one coil (Total of 4-wires)
- To switch the magnetic field (S-N), the applied voltage polarity must be switched so;
- Each wire of the coil needs to be connected GND or VCC (half H-bridge)
- The advantage gained in the simplicity of construction of the motor is lost in the control circuitry as it becomes more complex.
- Two H-bridges are required for the control of the motor.

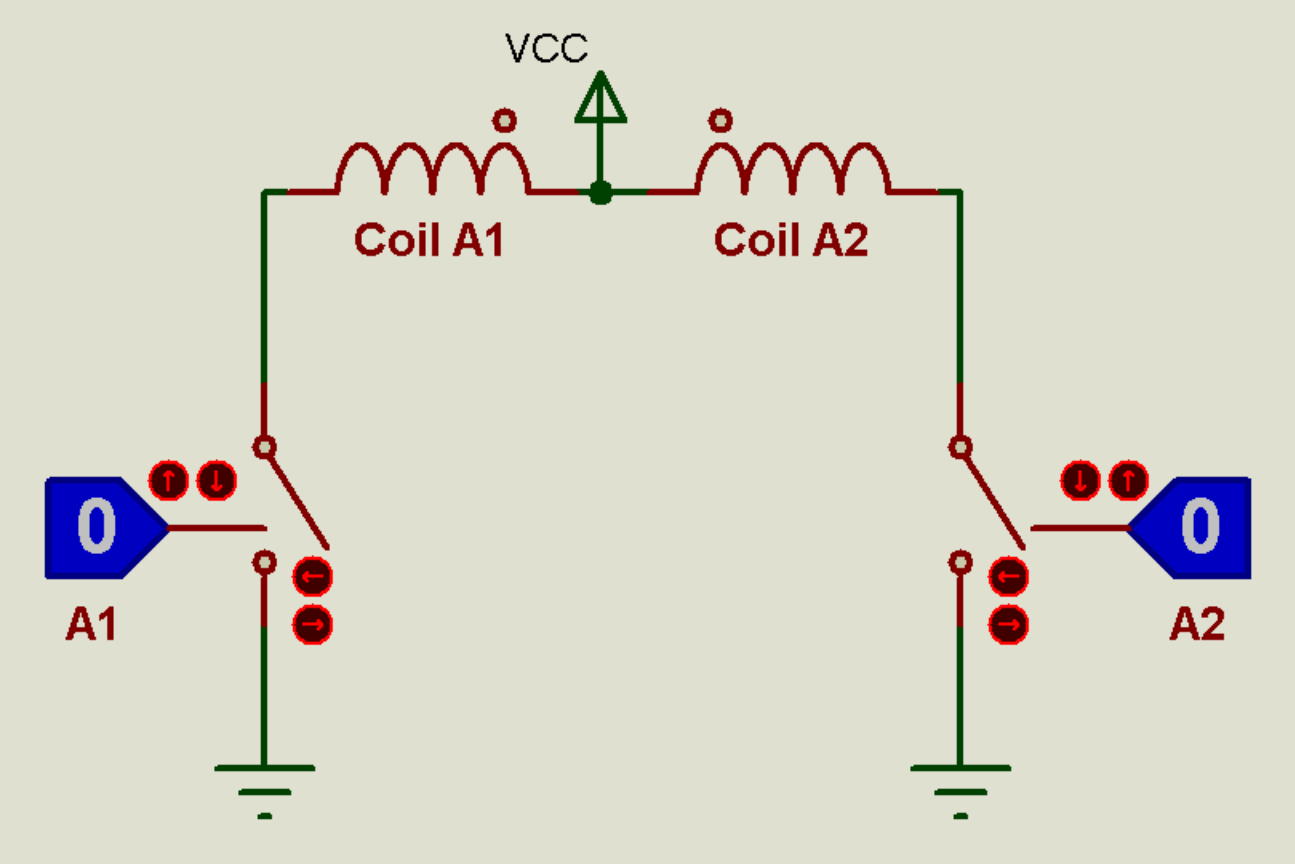
UNIPOLAR STEPPER MOTORS



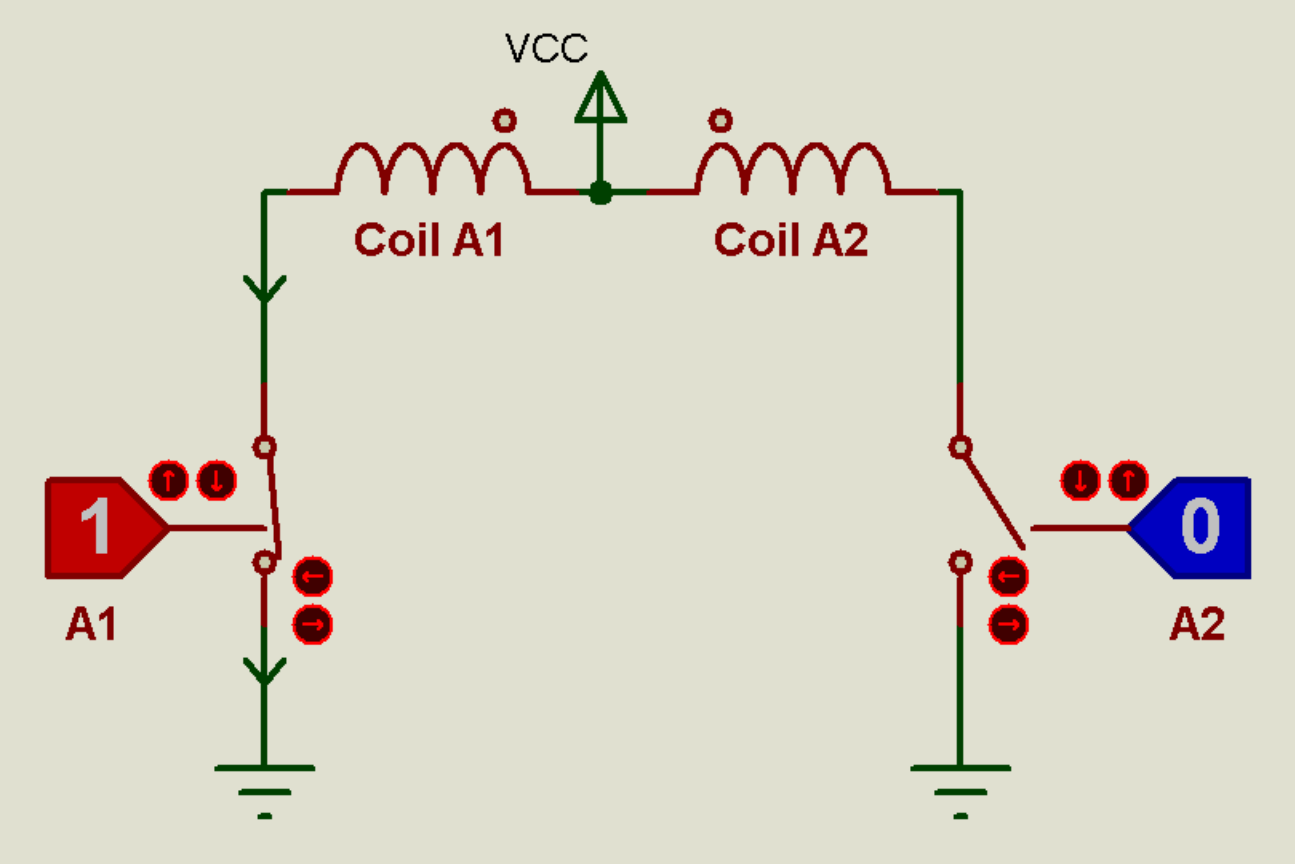


- Each pole consists of two coils (center tapped) → (Total of 6/5-wires)
- The center tap is connected to VCC
- To switch the magnetic field (S-N), GND is connected to either A or A-
- A simple switch is needed to connect coil pins to GND
- The advantage gained in the simplicity of the control circuit is lost in the complexity of motor construction

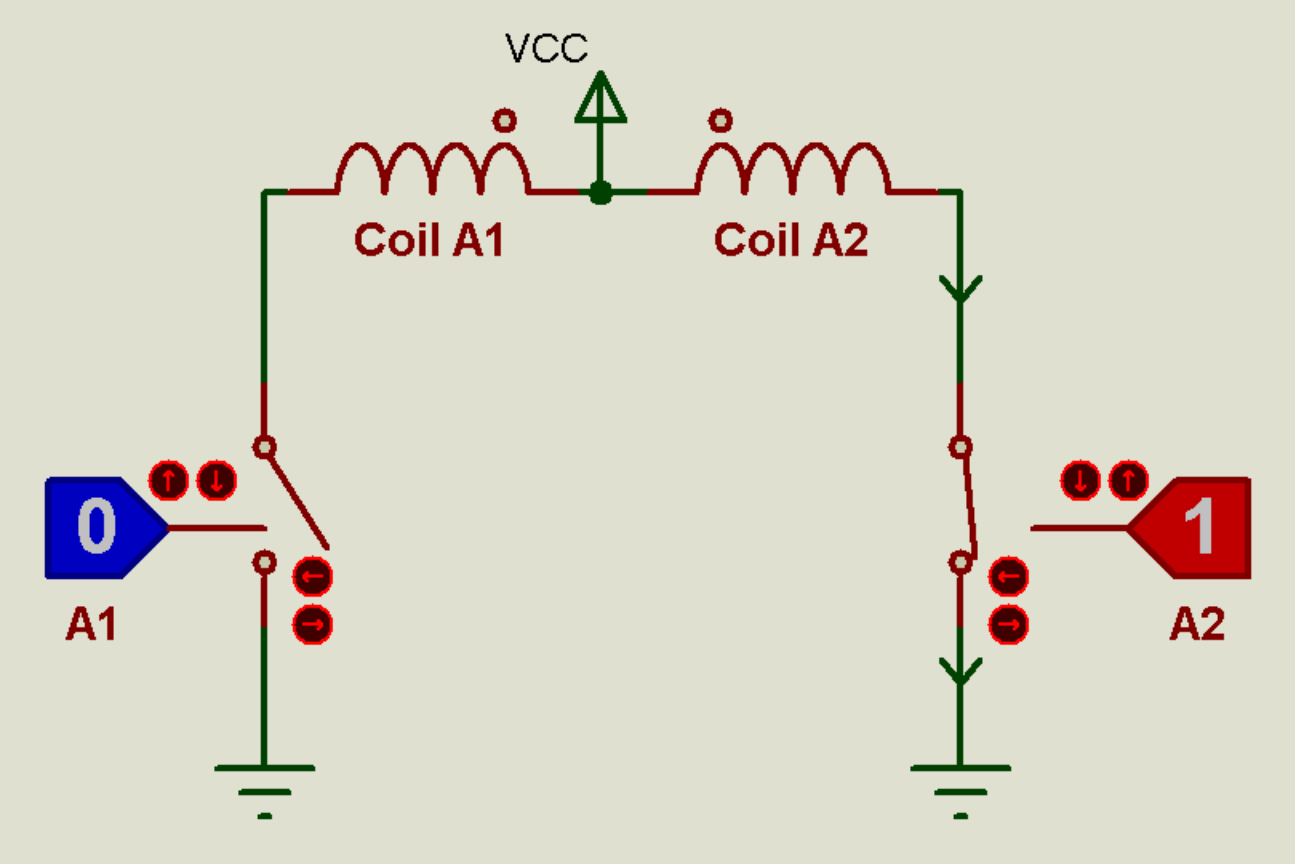
UNIPOLAR MOTORS



UNIPOLAR MOTORS



UNIPOLAR MOTORS



ADVANTAGES:-

- √ Their control is directly compatible with digital technology
- √ They can be operated open loop by counting steps, with an accuracy of ±1 step.
- √ They can be used as holding devices, since they exhibit a high holding torque when the rotor is stationary

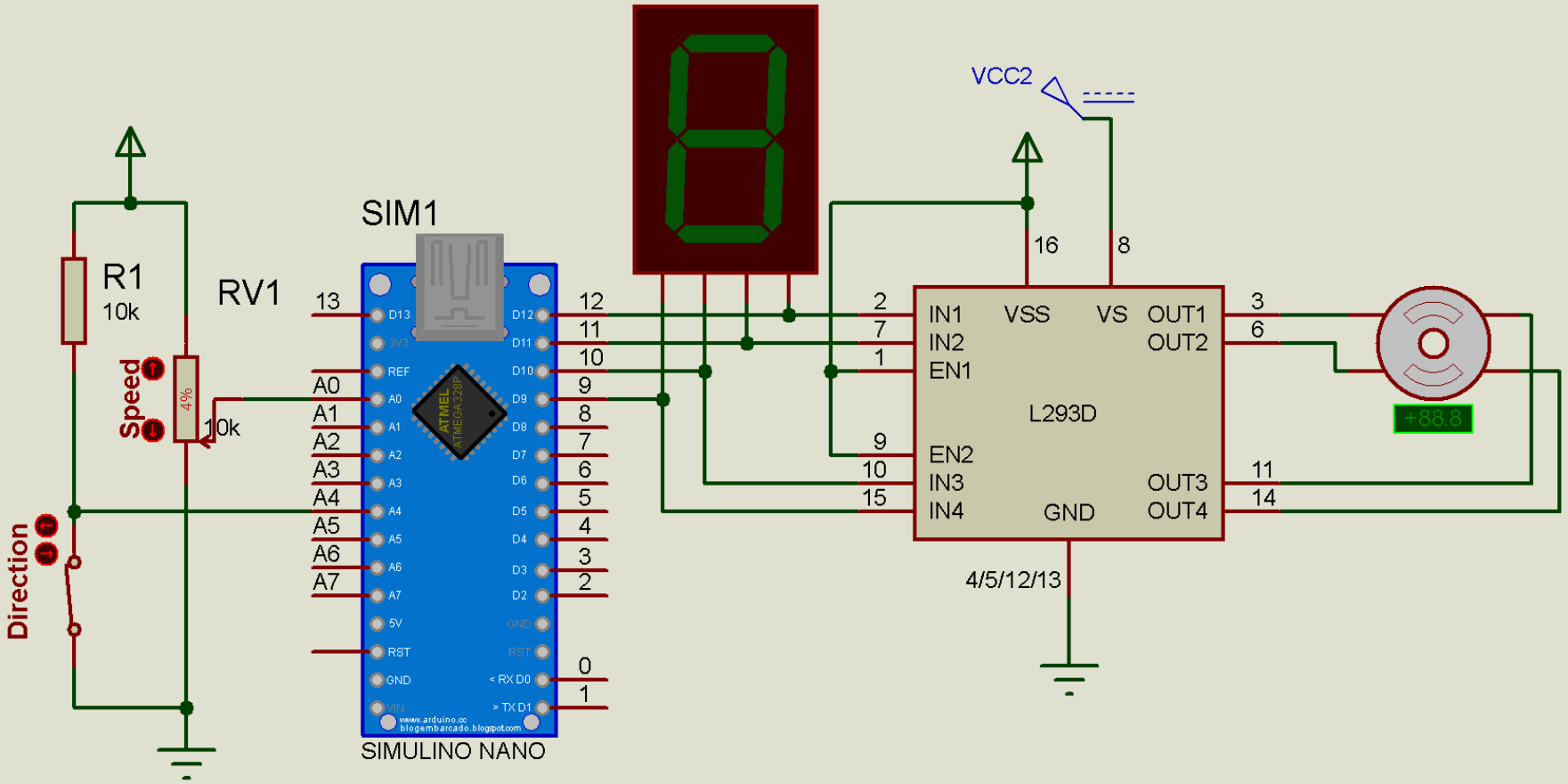


DISADVANTAGES

- Resonance effects and relatively long settling times
- Rough performance at low speed unless a micro step is used
- Liability to undetected position loss as a result of operating in open-loop
- Consume current regardless of load (coils are always energized)
- Losses at speed are relatively high and can cause excessive heating and they are frequently noisy (especially at high speeds)

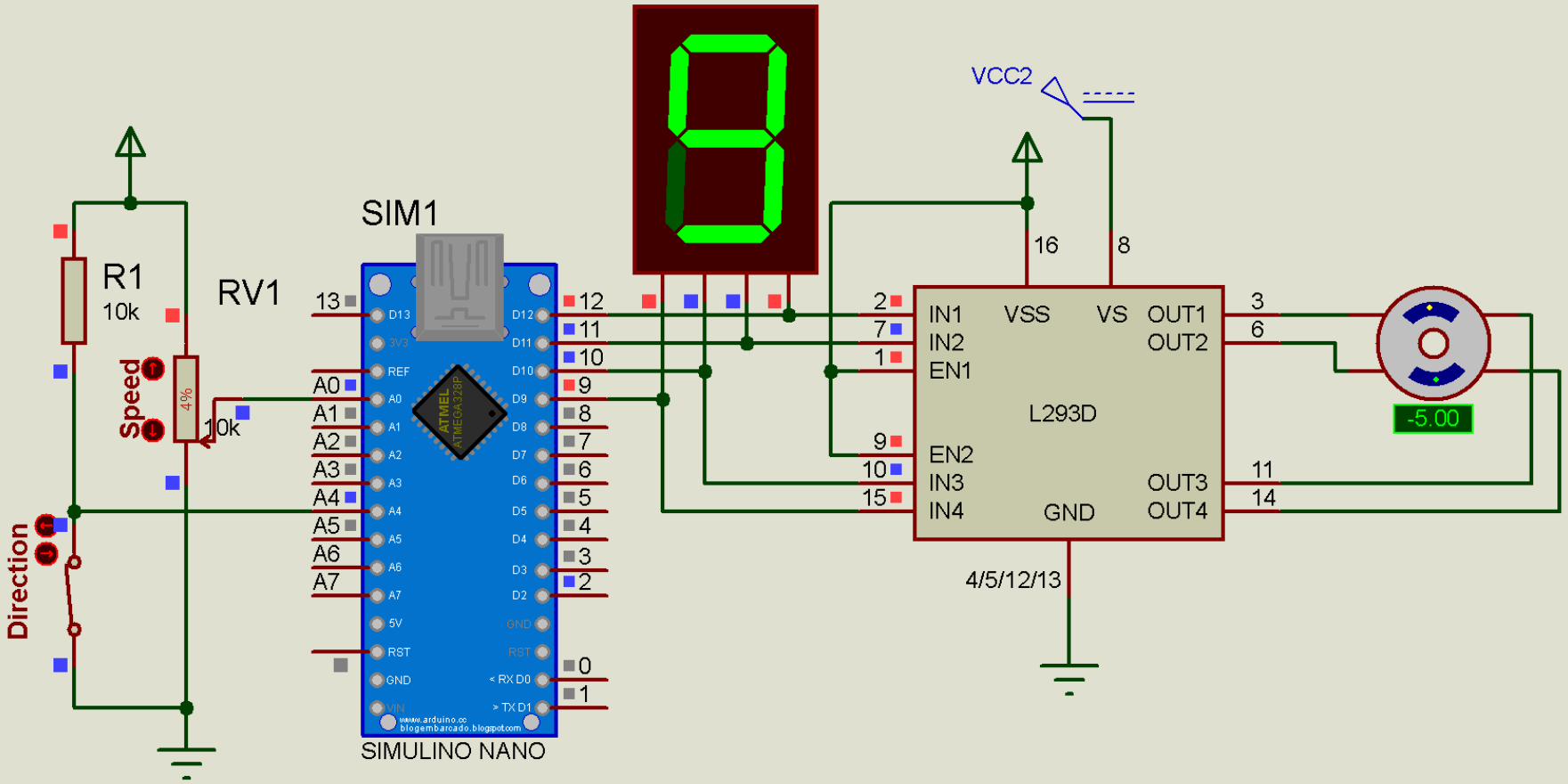


EXAMPLES BIPOLAR STEPPER

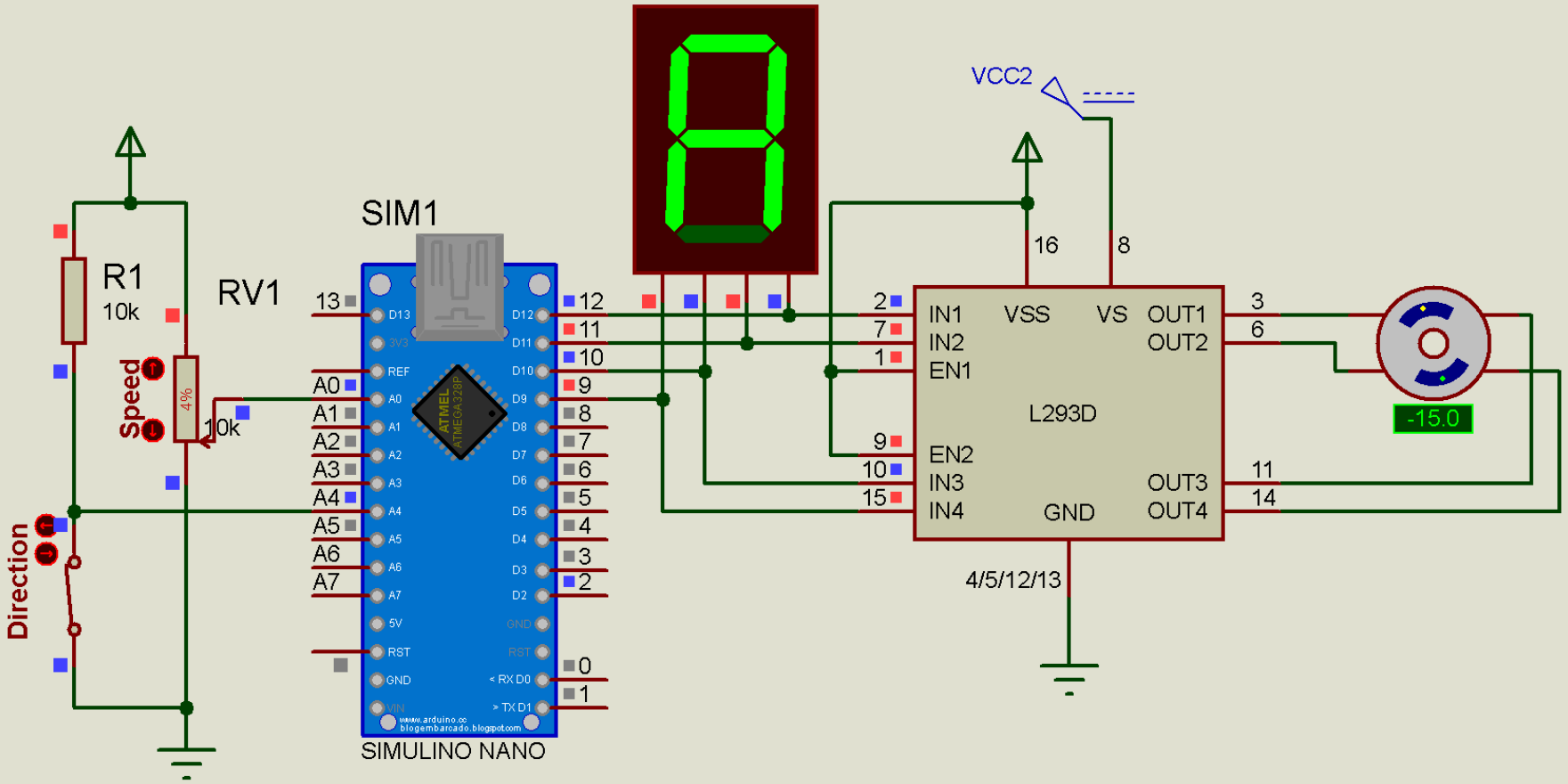


https://simple-circuit.com/arduino-bipolar-stepper-motor-control/

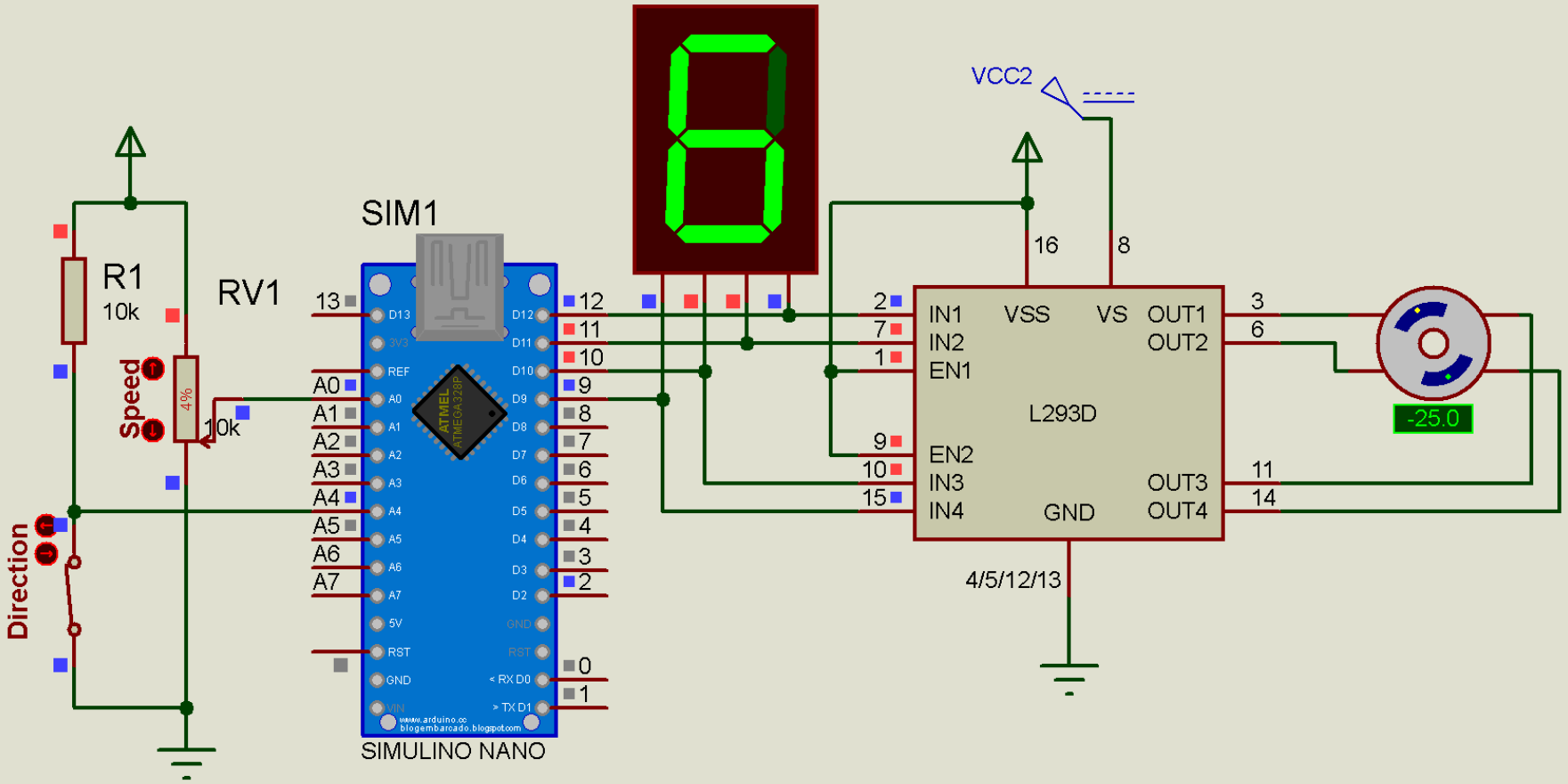
EXAMPLES BIPOLAR STEPPER



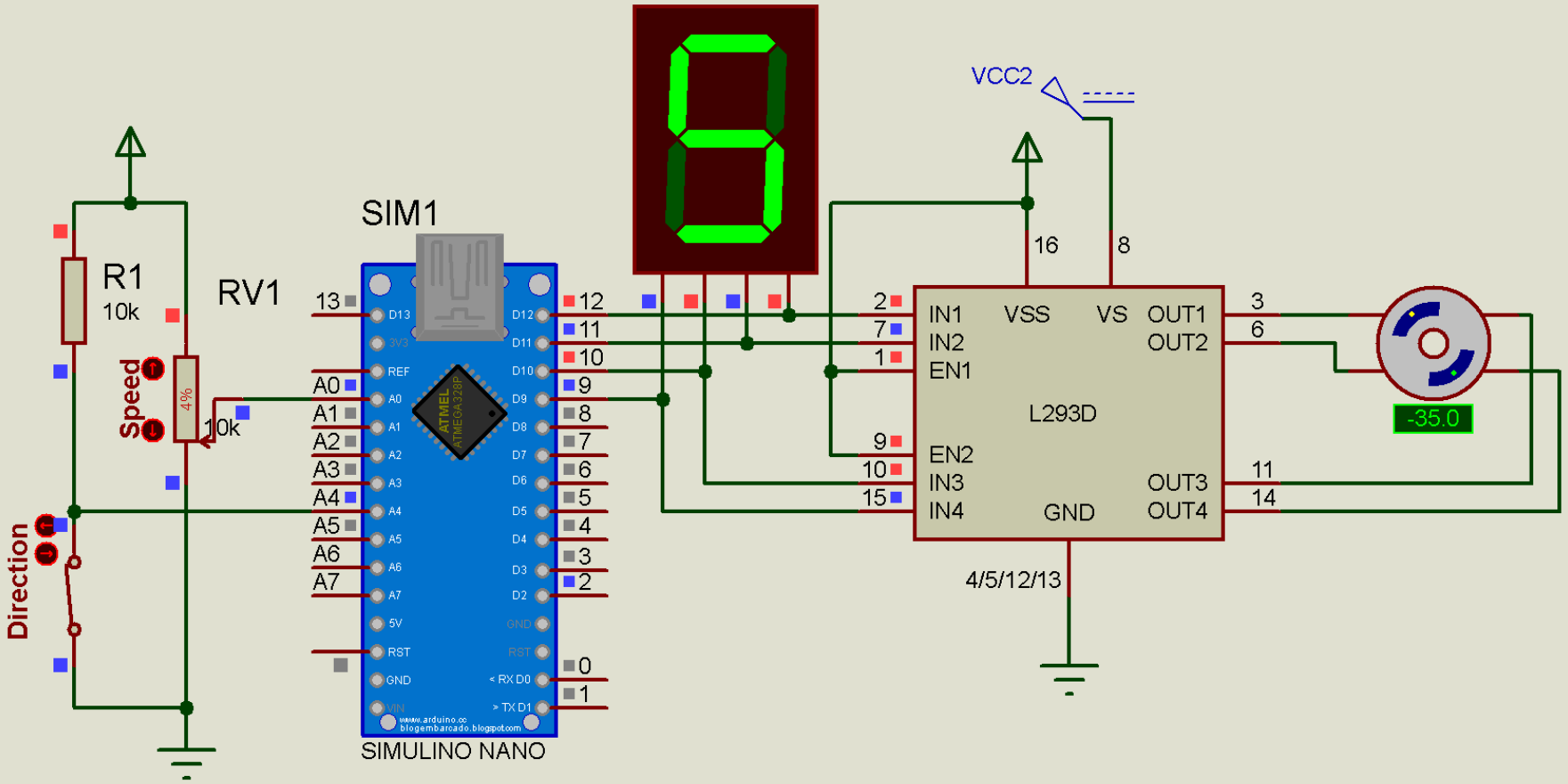
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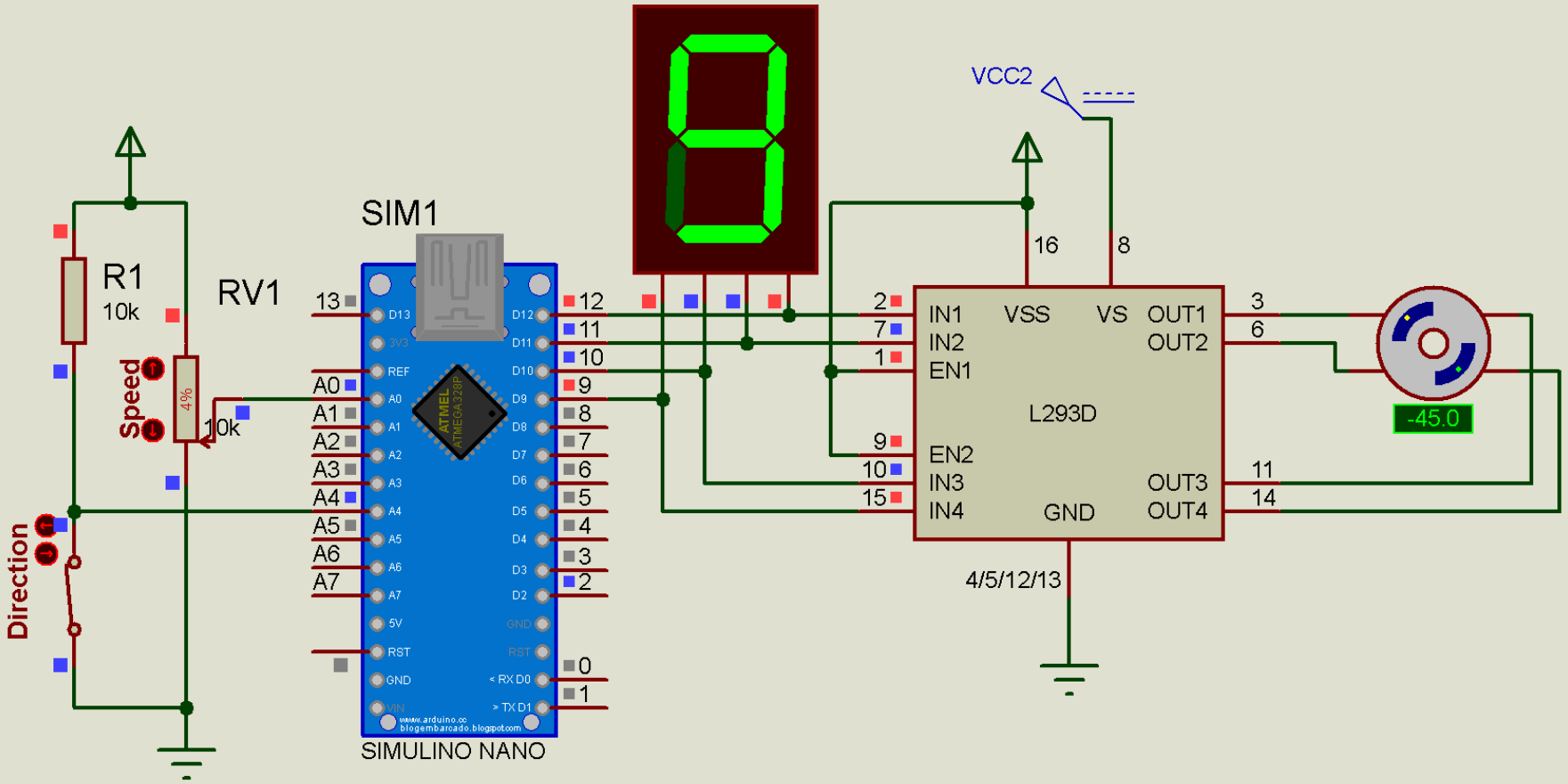
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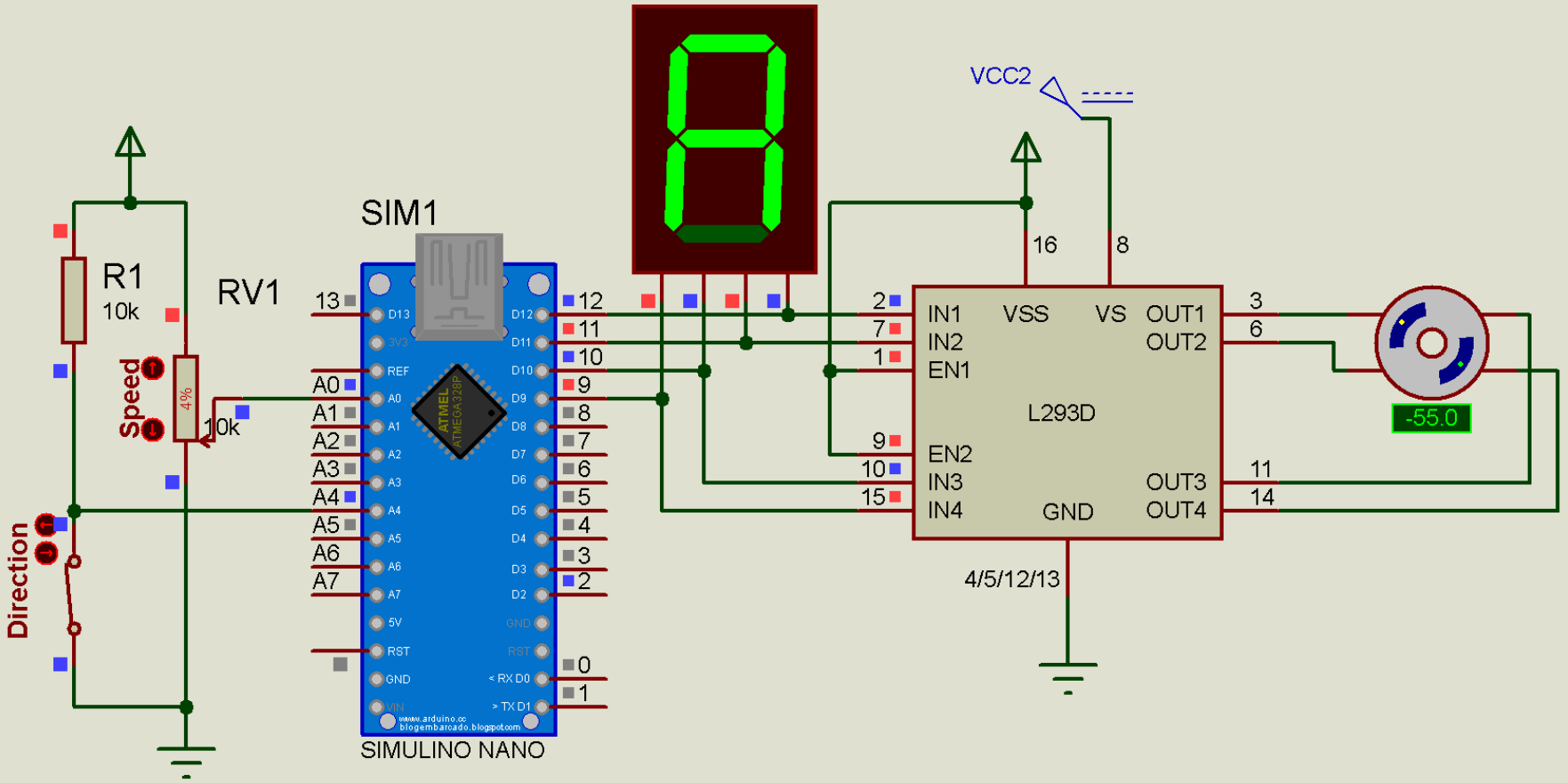
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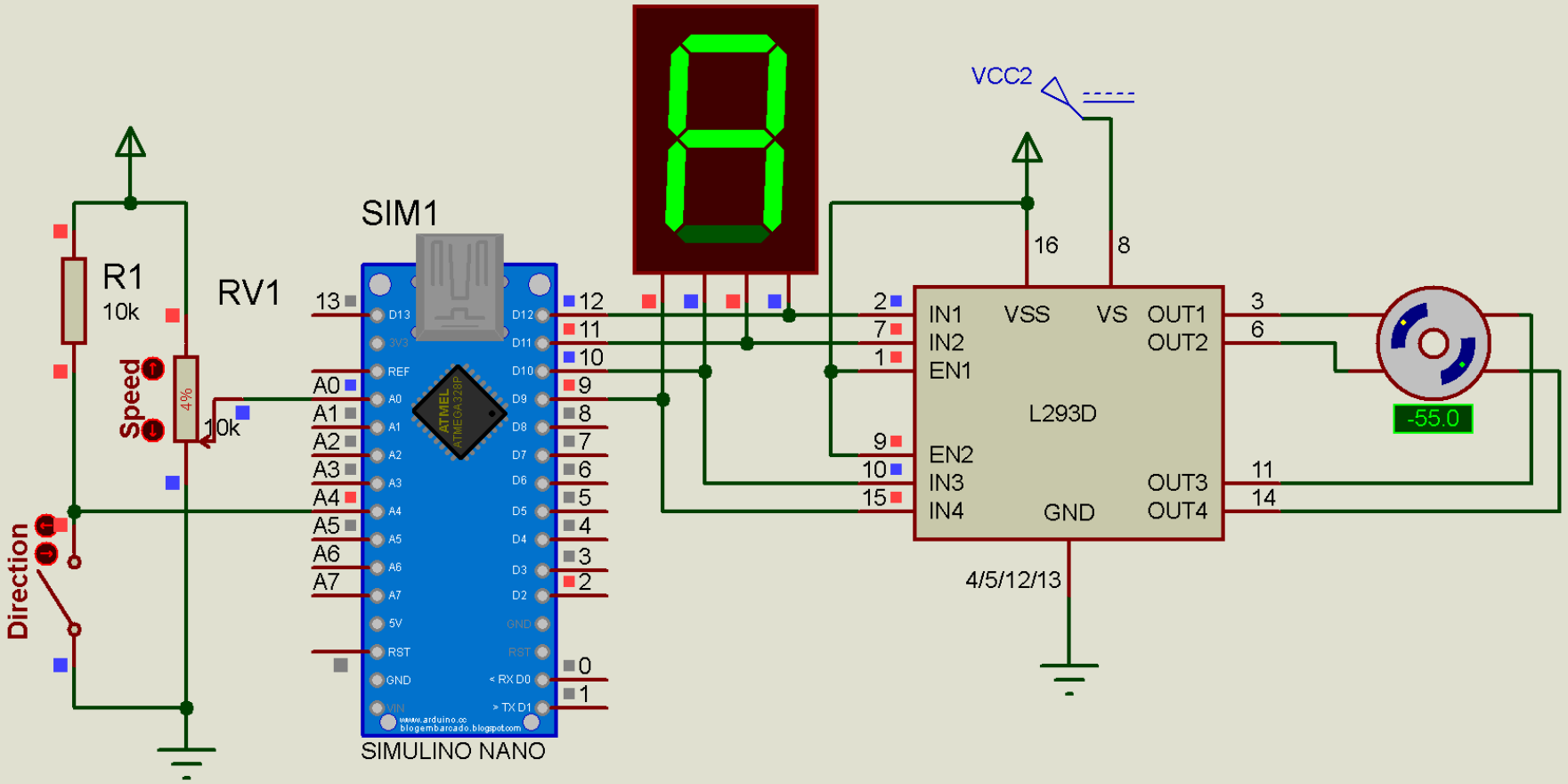
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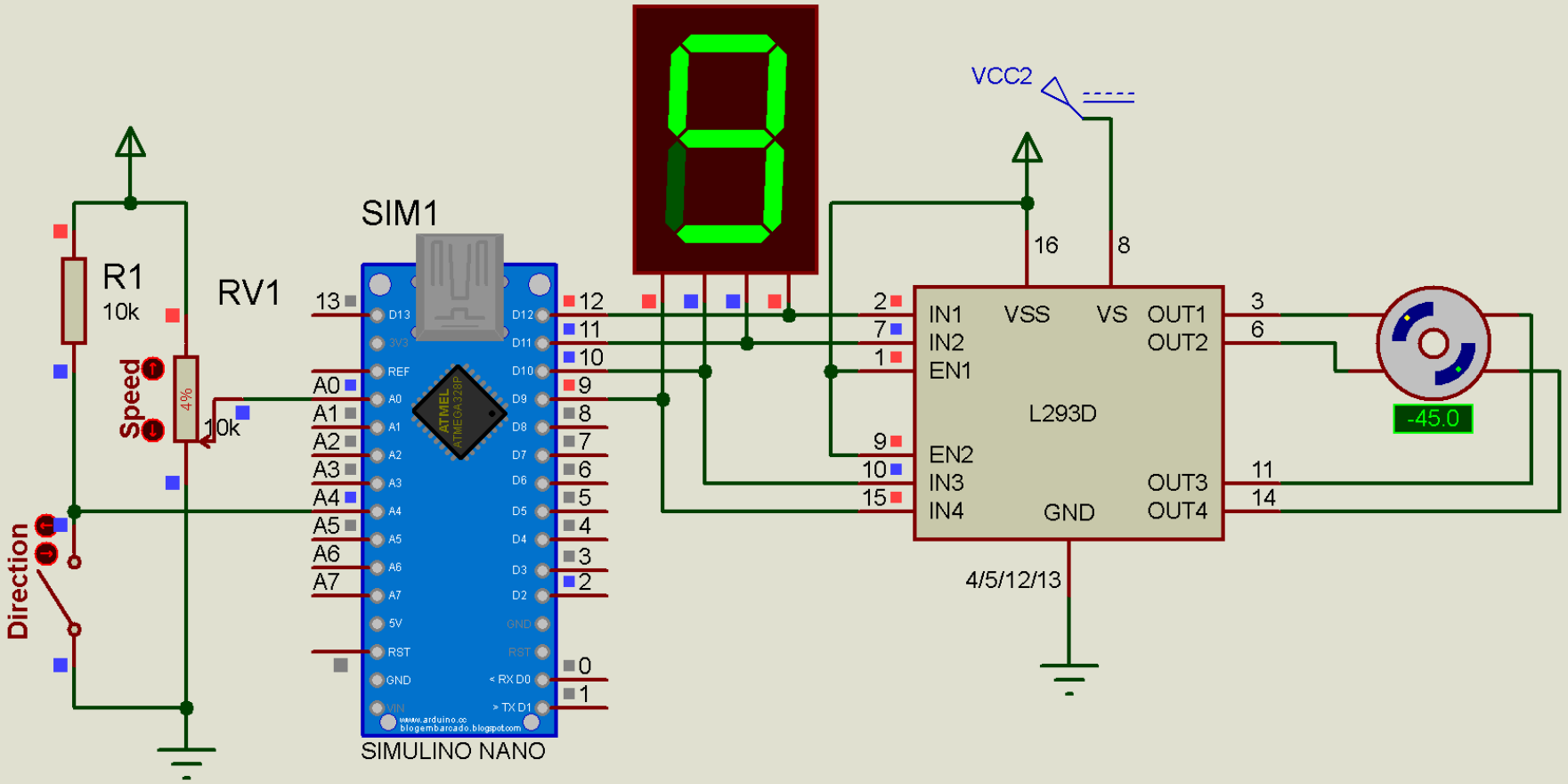
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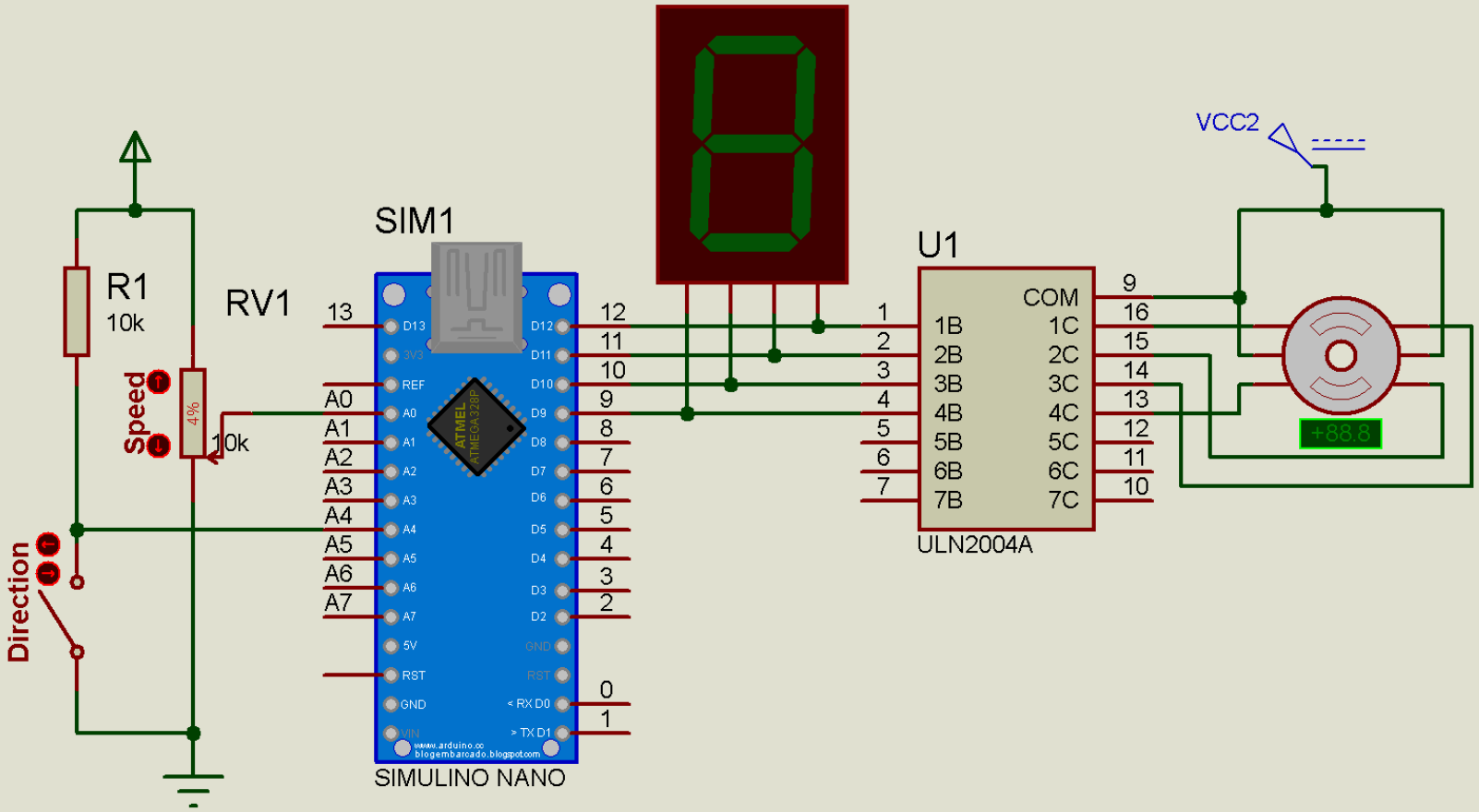
EXAMPLES BIPOLAR STEPPER



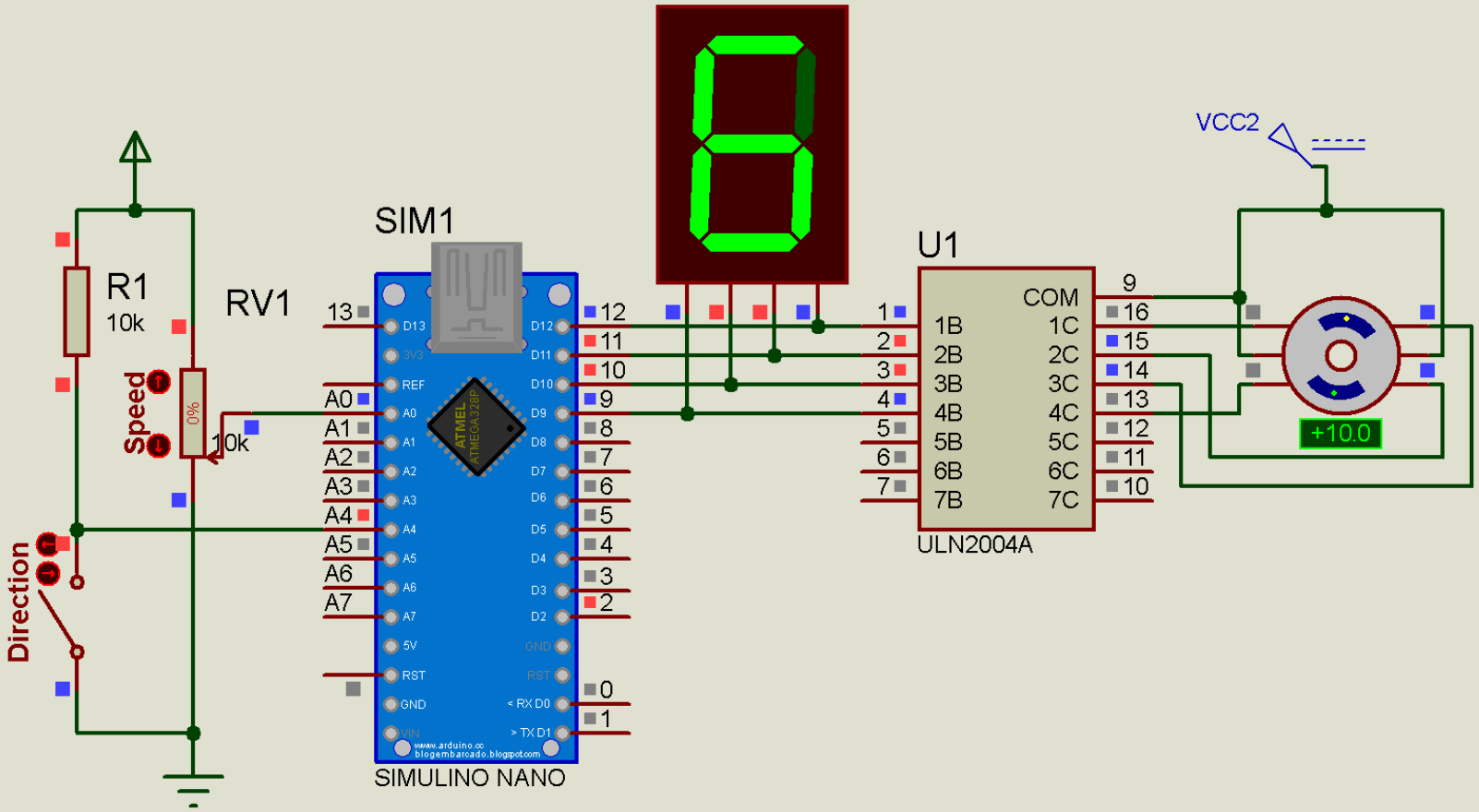
EXAMPLES BIPOLAR STEPPER



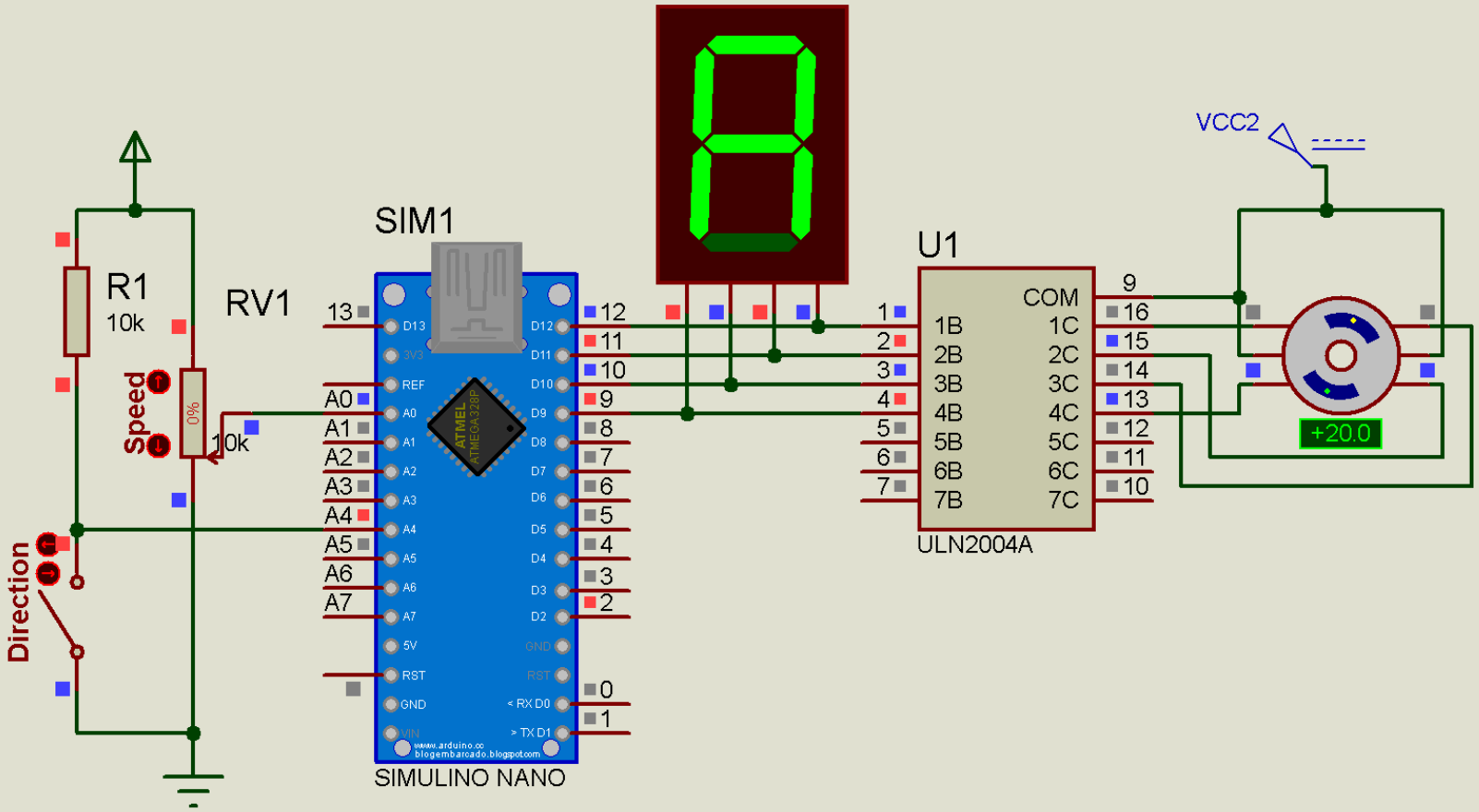
EXAMPLES UNIPOLAR STEPPER



EXAMPLES UNIPOLAR STEPPER



EXAMPLES UNIPOLAR STEPPER



THE END

